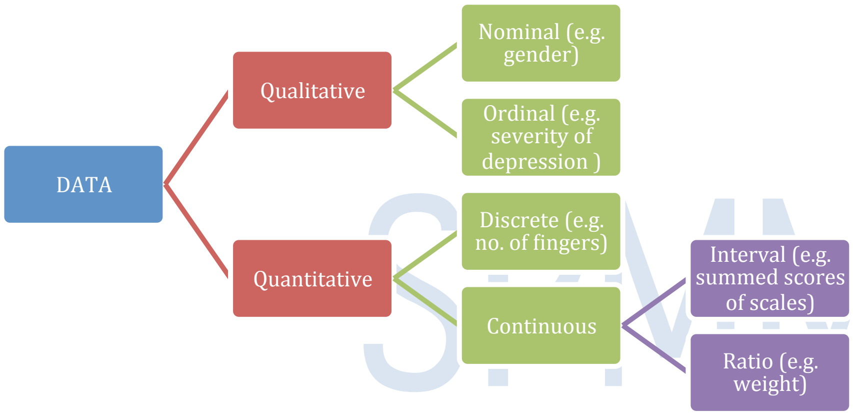
Variables & Data



The term “**Variable**” relates to anything that is measured or observed or manipulated in a study. Such measurements of variables yield data. The data collected in a study needs to be analysed, and how we do the analysis depends on what type of data it is.

Qualitative data arise when individuals may fall into separate classes. These classes generally do not have any numerical relationship with one another.

Nominal data are simply named categories. For e.g. sex of an individual – Male or Female. A nominal data with just two categories is called ***binary data***.

Ordinal data are a set of ordered categories. For e.g. consider the answers to an item in a questionnaire:

How anxious are you when you wake up in the morning?
1.Very high 2.High 3.Moderate 4.Slight 5.Not anxious

Though there is an order, one cannot say that the difference between 1 and 2 is the same as the difference between 3 and 4. Another example is the severity of depression: mild, moderate or severe.

Quantitative data are numerical, borne out of counting or measuring something.

If the values of the measurements are integers (whole numbers), then the data is said to be **discrete**. Generally discrete data are limited by the number of values they can have.

If the values of the measurements can take any number in a range, e.g. cholesterol levels, the data are said to be **continuous**. Continuous data can take an infinite number of possible values.

In fact, the discrete vs. continuous differentiation is not practically useful. Most continuous data can have only a finite set of possible measurements due to limitations of available instruments that measure them. So most discrete data can be treated on par with continuous data.

Interval data are measurements where the difference between two values is meaningful. For e.g. the difference between a temperature of 90 degrees and 80 degrees is the same difference as between 80 degrees and 70 degrees. But a ratio of 2 values may not be meaningful as there may not be a true zero point. For example, 20 degrees may not feel twice as hot as 10 degrees.

Ratio data have all the properties of an interval variable, and also has a clear definition of zero point, hence besides differences, ratios are also meaningful. Variables like height and weight are ratio variables. Acidity, expressed in pH, is not a ratio variable because a pH of 0.0 does not mean 'no acidity' – but it denotes 'some acidity'. A pH of 6 is not twice as acidic as pH 3.

In fact, the ratio vs. interval differentiation is almost never needed in social and medical research. Most rating scales have meaningful zero. So most interval data can be treated on par with ratio data.

Note that conversion to categories is possible for any quantitative scale e.g. IQ measurement in numbers such as 100, 120, etc. can be categorized to mild ,moderate or severe learning disability categories by specifying class intervals (0 to 30, 30 to 50, etc.).

Calculation	*Frequency (mode)*	*Median*	*Sum/Differences*	*Mean*	*Ratios (e.g RR)*
Nominal	√	X	X	X	X
Ordinal	√	√	X	X	X
Interval	√	√	√	√	X
Ratio	√	√	√	√	√

Ref: Norman & Streiner's PDQ statistics.

Types of variables:

	Independent values	Can be ranked	Can have meaningful average and units (mean / SD etc)	Has true zero value (so ratio meaningful)
Nominal	YES			
Ordinal	YES	YES		
Interval	YES	YES	YES	
Ratio	YES	YES	YES	YES

Independent variable – controlled by the experimenter, presumed to have a causal/antecedent effect on dependent variable. More than one independent variable can affect a single dependent variable.

Dependent variable – depends on the value of independent variable.

For example, in a study of smoking and lung cancer, smokers and non-smokers are followed up to see who develops more lung cancer. Independent variable is smoking; dependent variable is the development of lung cancer.

Mediator variable – it is not always possible to explain an association by using 2 variables such as A causes B. There may be C which mediates the effects of A in causing B – then C is called a mediator variable. The process is called *statistical mediation*. A complete mediator is one without which A cannot cause B. A *partial mediator* is one which when absent, reduces but not eliminates the effect of A on B. It is important that by definition a mediator must not cause an outcome directly on its own (i.e. C cannot cause B). Effect of mediators can be statistically analyzed using regression tests.

SPMM Memory Corner

Nominal and ordinal data are analysed using ***nonparametric tests*** while most measured variables are analysed using ***parametric tests***.

Describing data

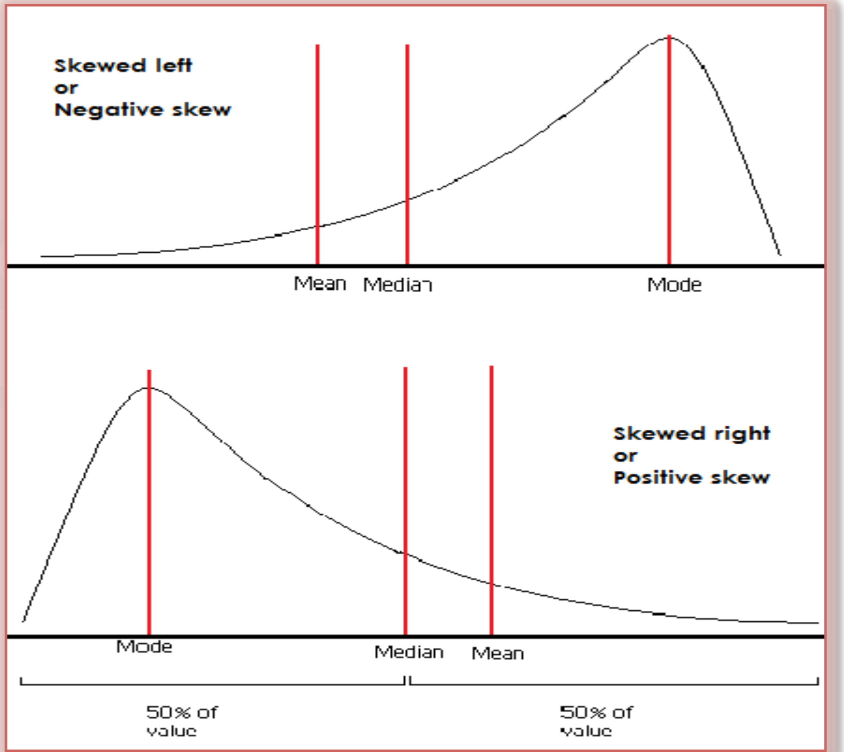
Descriptive statistics, as opposed to inferential statistics, helps to organize and summarize data in easily communicable manner. The purpose of description is not to make inferences about the sample, but only to enable observations to be meaningfully communicated, e.g. via the use of histograms, pie chart, and frequency polygons, etc.

Central tendency: the 'middle' of the collected data: Mean, median, and mode are measures of **central tendency.** The **mean** is simply the arithmetic average of the observed values. The mean is affected more by extreme values than is the median. The **median** divides the distribution into two equal groups. The median is a more stable indicator of central tendency than mean. The **mode** is the most commonly observed value in the distribution. There may be more than one mode in any given data set.

- ✓ **Mean**:
  - The sum of scores divided by number of scores (arithmetic mean).
  - Influenced by all available scores.
  - Highly sensitive to extreme scores i.e. easily influenced by an outlier value.
  - The higher the number of samples drawn, the more accurate the mean comes to the true population mean.
  - Mean can only be used for quantitative measures
  - The unit of means is the same as the unit of original measure.
  - The mean may be inappropriate for skewed data; it might be influenced by one way or the other. If individual observations are log transformed, then averaged, and then back-transformed using antilog we will obtain ***geometric mean***. This will be closer to median provided the log-transformed data had symmetrical distribution. This value will be less than the raw arithmetical mean.
  - The ***weighted mean*** is used when some individual observations are more or less valuable than others when reaching a summary measure. In the weighted mean, individual values are multiplied by weights (constants) attached to them before averaging.
- ✓ **Median:**
  - It is the point value that divides a distribution into two equal sized groups.
  - Half of scores fall below; half above.
  - It is also called as the 50^{th} percentile.
  - The median is not as influenced by extreme scores as mean, but it ignores most of the available information.
  - It is preferable for nominal data when treated as values (not as counts)

Percentiles & quartiles: Imagine a set of observations whose values are arranged in an ascending order. The value below which 5% of observations (i.e. 50 observations if the original set had 1000 individual values; 5 observations if the set contained 100 values) lay is called 5^{th} percentile. The value below which 95% of observations (i.e. 950 observations if the original set had 1000 individual values; 95 observations if the set contained 100 values) lay is called 95^{th} percentile. The value below which 50% of observations lay is called 50^{th} percentile or median. Similarly, if instead of 1 in 100 divisions ('per- cent – ile'), one uses 1 in 4 divisions, one may get quartiles. 25% values lay below 1^{st} quartile; 50% below 2^{nd} quartile (median) and 75% below 3^{rd} quartile. Statisticians often use the box-whisker plot in the context of quartiles.

✓ **Mode** :
- It is the most commonly occurring value in a distribution.
- It is a crude measure mostly used for nominal data (frequencies).
- It is also useful for ordinal data to understand the most common rating obtained on a Likert scale; but similar to the median, it ignores most of the available information.
- In bimodal distribution, two values occur equally frequently.



✓ **Skew:**
- In normal symmetric distribution, mean median and mode are equal. Distributions that are not normal are called **skewed distributions**.
- In **skewed distributions**, the extreme values will affect the mean to a larger degree than they affect median/mode. The mean will "chase the tail" of skewed distributions, whereas the mode will tend to stay home; median usually lies between mode and mean.
- In **positive skew**, higher extreme outliers are present, making the mean higher than median. Hence, the distribution curve shows longer tail on the right side.
- In **negative skew,** lower value outliers lead to mean being less than median and left tail being longer than right.
- Note that irrespective of the direction of a skew, the mode is always on the shorter side (outliers not as common as other values) and mean on the longer side.

Dispersion: the 'spread' of the collected data

- Measures of dispersion are also called as measures of variability and include range, variance, Standard Deviation (SD) and standard error (SE).
- **Range** refers to the difference between the highest and lowest scores in a distribution. It is easily determined when the data is arranged in a rank order (ascending or descending). It is very much distorted by extreme scores. It is better to present the minimum and maximum values along with range.
- **Interquartile range** refers to the difference between 75^{th} and 25^{th} percentile values. So 25% of the observed sample will be below the 1^{st} quartile, 25% above the 3^{rd} quartile while 50% between 1^{st} and the 3^{rd} quartile.

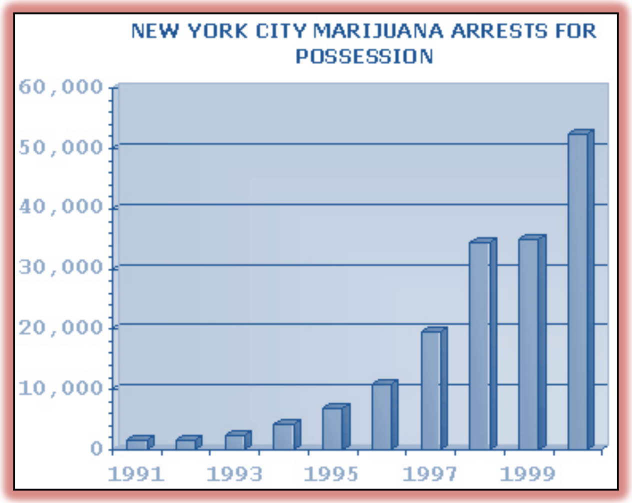
> **Variance =** Sum of squared differences of individual observations from mean / (number of observations-1)

- **Variance** is more informative than the range because it includes all scores in a distribution. When mean is subtracted from each individual score and the differences are squared and then summed followed by the division of this sum using N-1 (**degrees of freedom**), we can obtain a variance. Variance is high when scores are widely scattered. It becomes low when scores cluster around mean. Variance is expressed in squared units of the original measure.
- SD or **standard deviation** is the square root of the variance. It is the most commonly used measure of dispersion, largely because it has the same units as the primary data.

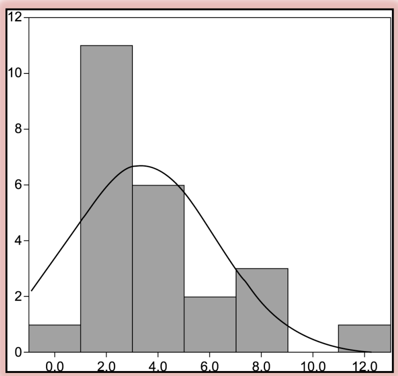
> **Standard Deviation = Square root of variance**

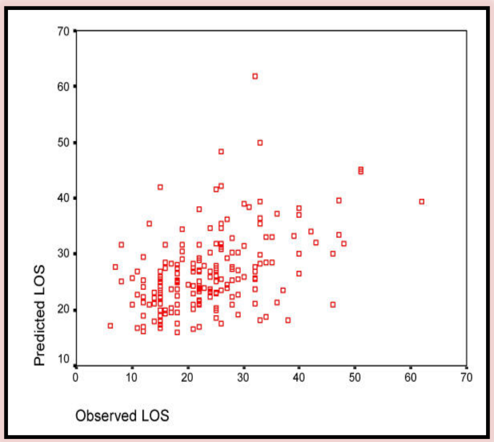
- **The coefficient of variation** is obtained by dividing the standard deviation by the mean and expressing this as a percentage. It is a measure of the 'relative' spread of the data.
- The **standard error of the mean** SE is the standard deviation divided by square root of sample size. So larger sample provides lesser SE and vice versa.
  - Consider a sampling distribution where individual scores are nothing but means of various samples drawn from a huge population. The variability of such a sampling distribution is expressed as the standard error. It is a better approximation of true population value that cannot be measured directly.
  - Authors often use the SE to describe the **variability** of their sample. SE gives an estimate of how the mean of the sample is related to the mean of the underlying population.
  - SE is always smaller than the SD. "Whereas the SD estimates the variability in the study sample; the SEM estimates the precision and uncertainty of how the study sample represents the underlying population. In other words, the SD tells us the distribution of individual data points around the mean, and the SEM informs us how precise our estimate of the mean is". (Retrieved from http://bja.oxfordjournals.org/content/90/4/514.full)

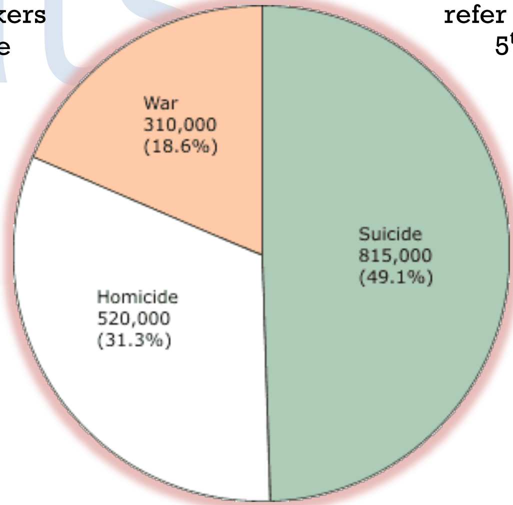
Displaying data as graphs/tables:



- An easy way of showing an observation to others is by obtaining frequency (counts) of observations in different categories. This is relatively straightforward to do for categorical data (e.g. count all females and males) and discrete numerical data (no overlap). We can use a ***bar chart or pie chart*** for this purpose.
- For continuous numerical data, something similar to bar chart is used – it is called a ***histogram***. Here there are no gaps between individual bars – as the data is continuous.
- Other options for continuous data include ***dot plot***, where a dot is placed for each observation along one axis.
- If a dot plot is extended to two axes, a scatter diagram is produced. In a ***scatter diagram***, two continuous measures can be plotted against each other.
- A ***stem & leaf plot*** involves plotting first few digits of numerical observations along a vertical axis, and then adding numbers to one (single group) or both sides (two groups) to represent individual values of observations.
- A ***box-whisker plot*** is, in essence, a rectangle drawn encompassing 2nd and 3rd quartile of observations. The median value can be shown as a line cutting through the rectangle. Whiskers refer to lines showing minimum and maximum values of the observation (range; one can also choose 5th and 95th percentile to display). Outliers can also be shown.
- We can also convert continuous data to categorical data and use pie/bar charts – but this means some useful information is lost.







Scatter-diagram: This is a study where correlation between 'observed length of stay' in psychiatric hospitals and 'predicted length of stay' using certain parameters for the same patients was analysed. Both variables were treated as continuous measures and a scatter gram was obtained.**BMC Health Serv Res. 2004; 4: 4.**

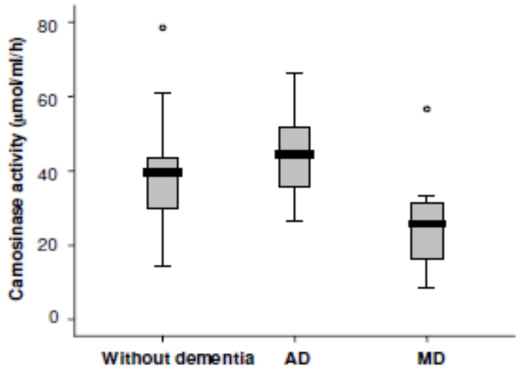


Figure 1
Box-and-whisker plot showing carnosinase activity in without dementia (n = 17), AD (n = 13), and MD (n = 7) groups. Comparisons were not significant between without dementia and AD or MD groups, however the difference between the AD and MD groups was significant (p = 0.01).

→ Box-whisker plot:

This is a study measuring enzyme carnosinase in those with Alzheimer's, mixed dementia and controls. Note the whiskers depicting lower and upper limits. The black horizontal line cutting rectangles show median (shifted upwards if outliers are present). The rectangle extends from end of 1st quartile to beginning of 4th quartile. The width of the box has no meaning. The dots are outliers **Balion, CM et al. Brain type carnosinase in dementia: a pilot study. BMC Neurology *2007, 7:38***

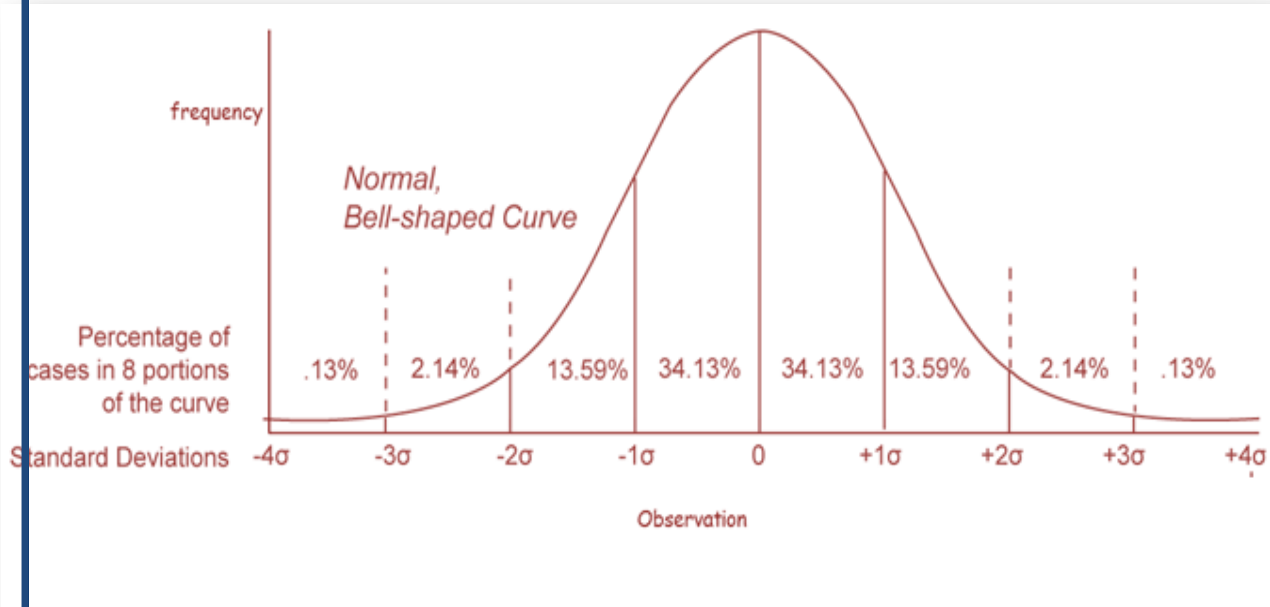
```
ISV                                    SPAQ

Freq.                                  Freq
 8     0 |00012334                     8     0 |00111111
14     0 |55666677888899               15    0 |222222233333333
24     1 |000000111122222333333444     22    0 |4444444444555555555555
20     1 |55566667777788889999         25    0 |6666666666666777777777777
22     2 |0000111122222333334444       25    0 |8888888888889999999999999
14     2 |55667788888899               12    1 |000001111111
15     3 |000011111233444              18    1 |222222223333333333
11     3 |55567777888                  11    1 |44455555555
 3     4 |124                           6    1 |666677
 2     4 |78                            3    1 |889
 3     5 |124
 1     5 |7
 1     6 |1
```

Stem & leaf plot: Seasonality of mood changes was measured in a sample of patients using two different scales – Seasonal Pattern Assessment Questionnaire (SPAQ) and Inventory for Seasonal Variation (ISV). The scores are depicted as a stem leaf plot. ISV stems are whole numbers and leafs are tenths of numbers, e.g. the lowest score is 0.0 and the highest score is 6.1. SPAQ stems are tens and leafs are ones; e.g. the lowest score is 00 and the highest score is 19. Note that the frequencies are indicated as a separate column.

Young, MA et al. Measuring seasonality: psychometric properties of the Seasonal Pattern Assessment Questionnaire and the Inventory for Seasonal Variation. Psychiatry Research 2003:117(1); 75-83

Normal distribution



Consider the "weights" of 100 psychiatry trainees appearing for paper 3. Once we have collected the data, you might want to plot a histogram of the observations on the x-axis and the frequency on the y-axis which shows how many times each weight occurs.

For a continuously distributed variable, provided the sample is big enough to represent the population, you can expect the data to be distributed symmetrically around the centre of all scores. If we drew a vertical line through the centre of the distribution, then it should look the same on both sides. This is called a **Normal distribution.**

Why is a "normal distribution" important?

1. A number of statistical tests assume that data come from a normal distribution.
2. In a normal population, the mean and variance (and hence SD) are not dependent on each other - i.e.
   a. 68% of the data will lie within 1 SD
   b. 95% will lie within 2 SD, and
   c. 99% will lie within 3 SD, irrespective of the mean.
3. Many natural phenomena are normally distributed – i.e. if we measure the height, weight, BP, etc. in a "large enough" sample, the frequency polygon would each approximately make a normal curve.

4. The **central limit theorem** states that if we draw equally sized samples from a non-normal distribution, the distribution of the means of these samples will still be normal, as long as the samples are "large enough".

Large enough – (depends on the hypothesis we are trying to test) - for most experimental purposes – 30 should be adequate to give a normal distribution.

Properties of a "normal distribution."

1. It is bell shaped
2. The mean, median and mode have the same value.
3. The curve is symmetric about the mean – i.e. the skew is 0.
4. The kurtosis (flatness of the curve) is also 0.
5. The tails of the curve reach close to the X axis, but never touches it.

To describe any normal distribution, two parameters have to be specified: the mean μ, where the peak of the density occurs, and the standard deviation (σ), which indicates the spread of the curve.

At a given value for a variance, the higher mean will shift the curve to the right; lesser mean will shift it leftwards. At a given value for the mean, higher SD will decrease the peakedness of the curve; lesser SD will increase the peakedness. The peakedness of such curves is called ***kurtosis***. A ***leptokurtic*** curve has a sharp peak. In other words, kurtosis is a measure of whether the data are peaked or flat relative to a normal distribution. That is, data sets with high kurtosis tend to have a distinct peak near the mean, decline rather rapidly, and have heavy tails. Data sets with low kurtosis tend to have a flat-top near the mean rather than a sharp peak. (Retrieved from `http://itl.nist.gov/div898/handbook/eda/section3/eda35b.htm)`

A **Standard Normal Distribution** refers to a normal distribution whose mean is zero, and SD is 1 unit. **Standard Normal deviate** is an expression denoted by z.

(Random value 'x' – mean)

z = -----------------------------

Standard deviation

SPMM Memory Corner

Normal distribution: Mean=median=mode

Positive skew: Mean>median

Negative skew: Mean<Median

A standardized Normal probability plot (z plot), is normal distribution with zero mean and standard deviation one

Standard error of the mean

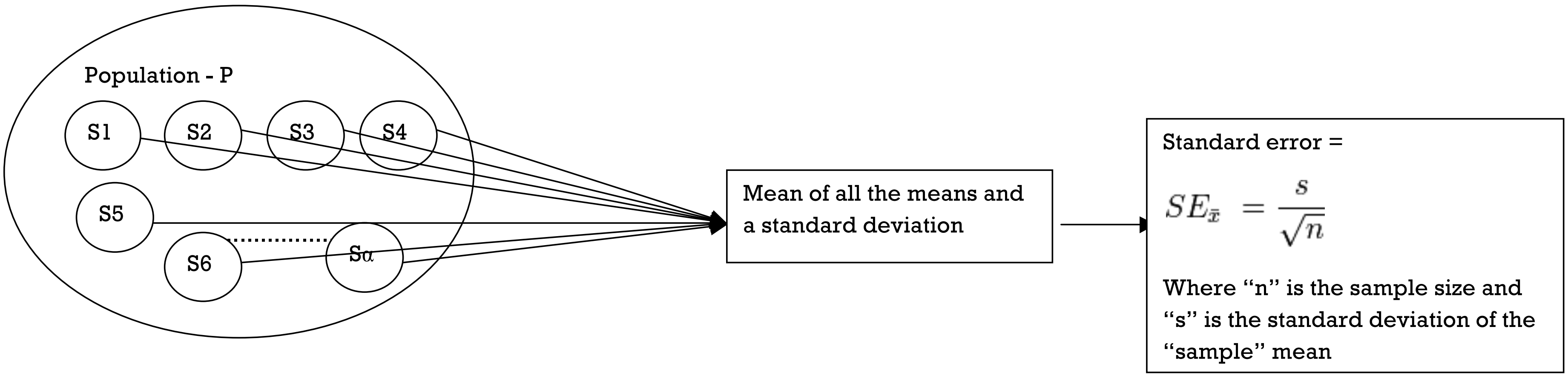
Calculated mean from a sample will never be exactly the same as the true mean value obtained by observing the whole population; this is called a random error, and it occurs by chance (not systematic). This error is represented by the standard error of the mean taken from that population.

Standard error of the mean (SEM):

Standard deviation refers to the dispersion of sample mean value around a range of observed values within a sample; standard ***error of mean*** refers to the dispersion of population mean value around a range of various mean values obtained from repeated sampling of the population.

In other words, the standard deviation of the distribution of the means is called standard error. i.e. if we take a number of independent "samples – S1, S2, S3 - Sα" from a "population P" – the standard deviation of the mean of the "means of each sample" is the "standard error" of the mean. Fortunately, we do not have to take "a number of independent samples" in order to calculate the SE. We can calculate the SE using the formula given below.

The standard error of the mean enables us to calculate the confidence intervals. Once we determine the SE, we can calculate the confidence intervals using various formulae



Odds and Probabilities

Both odds and probabilities are quantifications of the **likelihood** of an event. Although in everyday life we use the terms odds, probability, chance and likelihood interchangeably, these terms have specific meanings in statistics.

- **The probability** is the relative frequency of occurrence of a given event over a sequence of unlimited random trials done under similar conditions. In simple terms, probability compares the number of events of interest to the total number of trials during which the event may occur. It can take any value between 0 and 1 only. Probability can also be expressed in percentages (ranging between 0% to 100%).
    - Suppose 'p' is the probability of an event occurring. Then
    - If p = 0, then the event cannot occur
    - If p = 1, then the event must occur
    - The probability of an event not occurring is given by p = (1-p). This is called the probability of a complementary event.
- **Odds** are a ratio of the frequency that an event occurs to the frequency that the event does *not* occur (or a different event occurs in its place). In simple terms, odds compare the number of events of interest to the number of other events in the trial. So while calculating odds, we must keep the numerator and the denominator mutually exclusive.

Assume that a sequence of events makes up a total number N (e.g. number of coin tosses made). The events are of two types – event A (e.g. 'heads' in a coin toss) occurs n(A) times and event B (e.g. 'tails' in a coin toss) occurs n(B) times. Then,

P (heads) = n (heads)/No. of tosses

P (tails) = n (tails)/No. of tosses

Odds of heads = n (heads)/n (tails)

Odds of tails = n (tails)/n (heads)

Probability may be easily converted into odds using the following equation:

$$\boldsymbol{Odds = \frac{Probability}{1 - probability}}$$

Odds may be easily converted into probability using the following equation:

$$Probability = \frac{Odds}{1+Odds}$$

Probability of multiple events:

If two events X and Y are related but mutually exclusive (X can't happen if Y has happened) then probability of either X or Y happening is given by p(X or Y) = p(X) + p(Y). This is called the addition rule for mutually exclusive events. Examples of mutually exclusive events include toss of a coin - heads and tails, or blood groups A, B, O and AB. Probability of getting a head or a tail for a toss is P(Heads or Tails) = P(heads) + P(Tails)

If two events X and Y are unrelated then probability of both X and Y happening is given by p(X and Y) = p(X) * p(Y). This is called the multiplication rule for independent events. Consider the probability of finding a male with blood group A. Being a male does not affect the blood group – hence these two are independent events. The probability is given by P(Male with A group) = P(Male) X P(Blood group A)

(There are more addition and multiplication rules, but the above must suffice for the Membership exams.)

SPMM Memory Corner

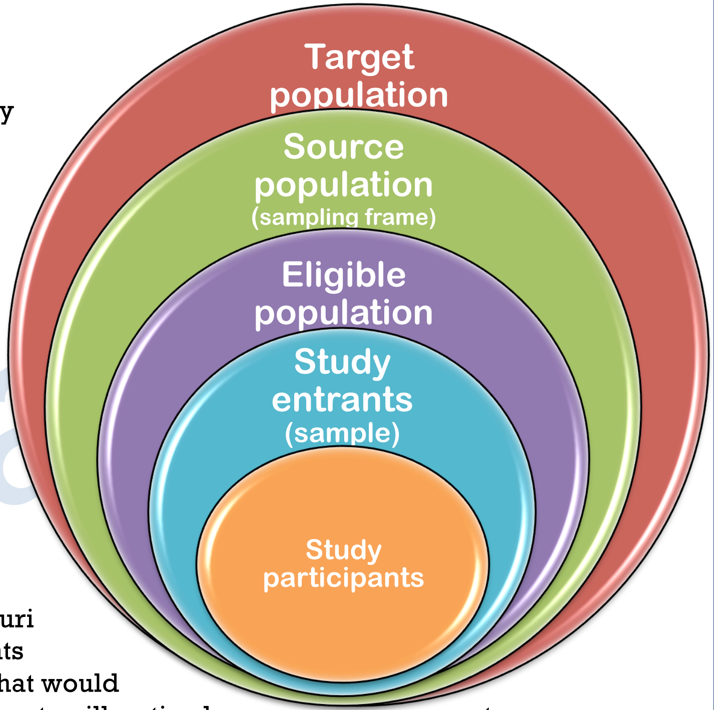
P (A) = n (A)/N
P (B) = n (B)/N

Odds of A = n (A)/n (B)
Odds of B = n (B)/n (A)

Population & Sample

Target population: The population to which we aim to apply our results (research is 'targeted' to the benefit of this group)

Source population: This is the **sampling frame** – a list of accessible members of the target population.



Eligible population: A large fraction of the source population will be excluded when the study criteria are applied. Those subjects who are not assessed for the study will also be ineligible.

Study entrants: When the eligible population is approached to take part in the study, a large portion will be excluded due to various issues such as lack of consent, confidentiality issues, administrative issues and non-response. Very few from the eligible population generally agree and enter a study.

Study participants / completers: Not all who enter a study will complete study requirements or be available for follow-up. The remnants that satisfy all study requirements and complete them are in this group.

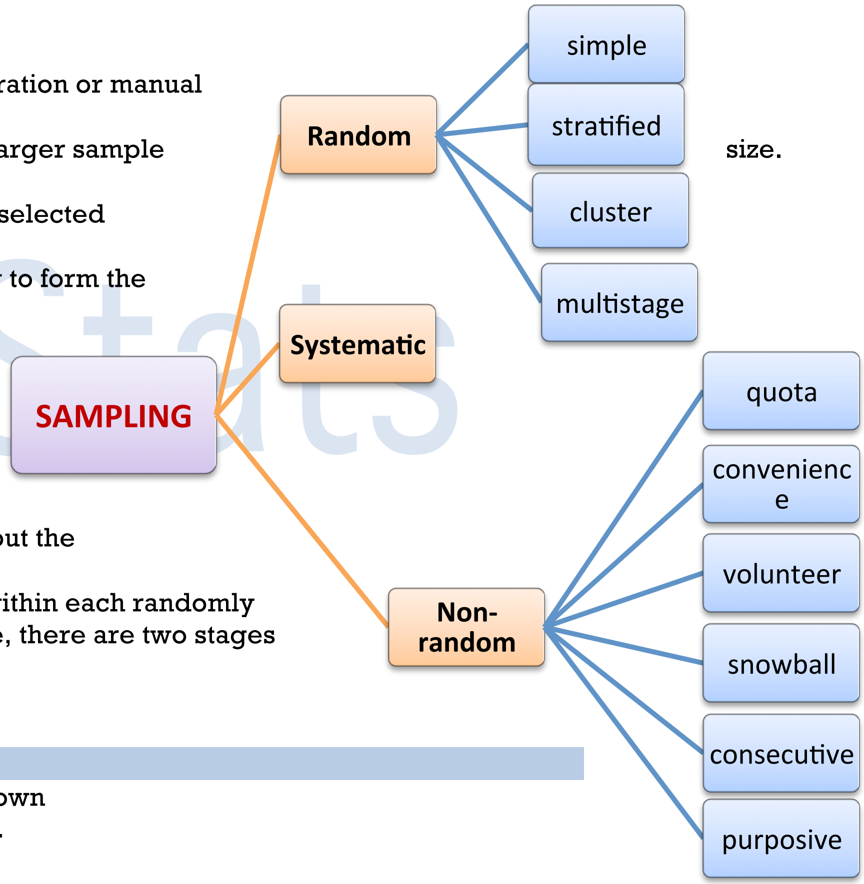
Consider the following study (Freedland, KE, et al. *Psychosomatic Medicine* 65:119-128 (2003)). A research team was interested in studying the prevalence estimates of depression in hospitalized patients with congestive heart failure (CHF). Patients age 40 years or older hospitalized at Barnes-Jewish Hospital, Washington University Medical Center, St. Louis, Missouri were included in the study. Of nearly 3900 eligible patients that were approached, 1900 patients were excluded due to dementia, delirium, or other neurological or psychiatric complications that would preclude valid assessment of depression or due to other reasons such as patient refusal (' feeling too ill or tired to participate') etc.

Here, 'population' = all hospitalized patients with CHF
'Sampling frame' = Inpatient registers at Barnes-Jewish Hospital, Washington University Medical Center, St. Louis, Missouri.
'Sample' = 2000 patients that met inclusion and exclusion criteria.

Sampling error refers to the difference between the outcome value estimated from a sample and the true whole-population outcome one might get by including every individual in a population. This can be measured. Please note hat this is different from systematic error or bias (discussed elsewhere). It depends on 1. Sampling technique. (Random sampling reduces error). 2. Sample size. (Larger samples produce less error)

Random sampling:

- Every individual in the population has an equal chance of being chosen.
- Avoids selection bias.
  - **Simple random sampling:**
    - Uses a set of randomly generated numbers using computer generation or manual draw.
    - It is simple and easy for small samples but may get difficult with larger sample size.
  - **Stratified random sampling:**
    - Sampling frame is divided into subgroups called strata based on selected variables
    - Then a random sample is picked up from each stratum separately to form the required study sample
  - **Cluster random sampling:**
    - The sampling unit is not the individual study subject but a pre-existing group e.g. PCTs. Counties, districts, etc
    - Once a cluster is selected randomly from a list of clusters (sampling frame), all subjects within that cluster will become study subjects
    - Cluster sampling can reduce costs and the amount of fieldwork, but the power of a study may be compromised.
  - **Multi-stage sampling**: Initially clusters are selected randomly. Then within each randomly selected cluster, a random sub-sample of individuals are picked. Hence, there are two stages of random sampling.



Non-random Sampling:

- The chance of a member of the population being in the sample is unknown
- Generalisability is weaker in studies employing non-random sampling.
- Also called non-probability sampling
  - **Quota sampling**:

- Sampling frame is divided into subgroups called strata based on selected variables
- Then a non-random sample is picked up from each stratum separately to form the required study sample e.g. first 10 subjects in each stratum etc.

· **Convenience sampling**:
- Subjects are selected according to their availability and easiness of access e.g. first few patients in an outpatient list.

· **Volunteer sampling**:
- All those who volunteer for the study are included.
- Note that voluntary respondents may differ from non-respondents.

· **Snowball sampling**:
- Used when the desired sample is difficult to reach e.g. drug users.
- One identified subject helps to locate another similar subject.

· **Consecutive sampling:**
- A case series of consecutive patients with a condition of interest.

· ALL patients that present to the research site at the time of study will be included.

· **Purposive sampling**:
- Often used in qualitative research.
- Study subjects are deliberately selected according to various pre-requisite objectives.
- **Includes**
  - **Extreme sampling** wherein cases representing two extremes of the phenomenon under study are recruited.
  - **Critical case sampling** wherein subjects who are expected to yield the maximum information are selected
  - **Theoretical purposive sampling** wherein subsequent recruitment depends on those categories that emerged from data collected so far.

Systematic sampling:

✶ This is also called as **interval sampling**.
✶ Every n^{th} item from a sampling frame is selected as the study subject.
✶ Strictly speaking, it is not a random sampling technique though some may argue against this.

Sampling techniques – key facts:	**Snowball sampling:** asking individuals studied to provide references to others; may be useful in identifying sexually transmitted disease samples, IV drug users samples etc.
Cluster sampling: dividing the population into clusters, typically on the basis of geography, and taking a sample of the clusters.	**Random sampling:** every individual in population has an equal chance of being chosen.
Convenience sampling: selecting on the basis of convenience – not using a sampling frame but including whoever is easily accessible.	**Stratified sampling:** dividing the population into groups on the basis of some suspected confounding characteristic and then sampling each group to ensure fair distribution of these variables.
Quota sampling: selecting fixed numbers of units in each of a number of categories.	**Systematic sampling:** choosing every nth item from a list, beginning at a random point.

Inferential statistics

We previously saw some methods that help to describe the data collected from a sample (descriptive statistics). Often in research, we need to go a step further than description: making generalisations or inferences. Inferential statistics enable us to make inferences about the nature of a population using the data from a sample. Statistical inference may take the form of:

1. **Estimation**: Using the sample data we estimate the distribution of a parameter in the population from which the sample was drawn. There are two types of estimation: a**. Point Estimation**: estimate a single value for a parameter that will probably be close to the true value of the parameter (effect sizes). b. **Interval Estimation**: find an interval (confidence interval) that has a given probability (the confidence coefficient) of including the true value of the parameter within its specified range.

2. **Hypothesis Testing**: We test the null hypothesis that a specified parameter of the population has a specified value by looking at the sample's value.

Hypothesis testing:

- *Hypotheses* are conjectural statements that provisionally link two variables. In contrast, *theories* are sets of definite propositions or facts that are more or less verified already. Stating a hypothesis linking two variables is often the first step in examining the relationship statistically using probability theory.
- *According to Popper's logic, to prove something is extremely hard. But to disprove is relatively easy. Hence, science does not use the method of verification: but uses the methods of falsifiability. Hence instead of attempting to prove something is correct, researchers often go on to disprove that the opposite is incorrect.*
- **Null hypothesis**: H0 – This is the statement in opposite which scientists want to disprove (or reject). If you want to show that A and B are different from each other, then the null hypothesis states that no difference exists between the two groups A and B. All statistical methods that follow this try to refute this statement using statistical inferences.
- **Alternate hypothesis** or H1 gets accepted if the null hypothesis gets correctly rejected. The alternate hypothesis is what the scientists actually believe in and try to demonstrate as true to the scientific world. The alternative hypotheses (H_1) is the antithetical complement of the H_0. The alternate hypothesis is often the same as the original research hypothesis.
- There are two ways of stating a hypothesis: one-tailed or two-tailed.

- **One-tailed hypothesis** refers to the statement that differences between groups occur in one direction only i.e. A >B. In this case, the alternative hypothesis will be A is not > B.

- **Two-tailed hypothesis** refers to the statement that differences exist between the two groups but the direction of such differences are not specified i.e. it may either be A > B or B > A but importantly A ≠ B. In this case alternative hypothesis will say that A = B. Significance levels of two-tailed hypothesis are halved – so it is necessary to have larger differences to reject a two-tailed null hypothesis. This is considered to be more rigorous and so often preferred in clinical trials.

The Null Hypothesis H0 is tested by gathering data that are relevant to the hypothesis and determining how well the data fit the H_0. If the fit is poor, we reject the H_0 and approve the H_1 (this is desirable for the researcher). When we test how our data fits with H_0, we use a significance level, p, which is the probability of rejecting the H_0 (and approving H_1), when H_0 is indeed true. The higher this p, the better the fit between the data and the H_0. If this p value is low, we have cast doubt upon the validity of H_0. If p is very low, we can reject the H_0 (this adds support to the researcher's original assumption). Arbitrarily the notation 'very low' is taken as a p value of 0 .05.

Errors:

Random errors are fluctuations in either direction in the measured data due to the precision limitations of the measurement device. This is often due to researcher's inability to take a measurement in the same way to get exactly the same result. **Systematic errors** are also called bias. These are reproducible errors that are consistently in the same direction. A problem which persists throughout the entire experiment leads to bias. These are dealt along with study designs in further detail. Two very different league of errors can occur during hypothesis testing.

Type 1 error:

- Incorrect rejection of the null hypothesis (**false positive** claim in favour of the research hypothesis).

- Consider an overzealous SpR undertaking a research project. He might uphold the alternate hypothesis and reject the null one in error – this is called type 1 error.

- The likelihood of type 1 error is called **alpha** (1 for alpha) – and as long as it is less than 0.05, it is well and good for most cases. This is said to be the level of **statistical significance** (p).

- Repeated testing of hypothesis using a same data, multiple subset analysis and secondary analysis can lead to type 1 error as in these cases at least one test will be positive out of 20 if 'p' is set at 0.05. Read further for correction of this error.

Type 2 error:

- **It** is the incorrect acceptance of the null hypothesis i.e. concluding that no differences exist between two treatment groups when they actually exist (**false negative** rejection of research hypothesis).
- The likelihood of type 2 error is called **beta**.
- This is mostly due to small *sample size* or *large variance*.
- 1-beta refers to the **power** of a study. (see below)

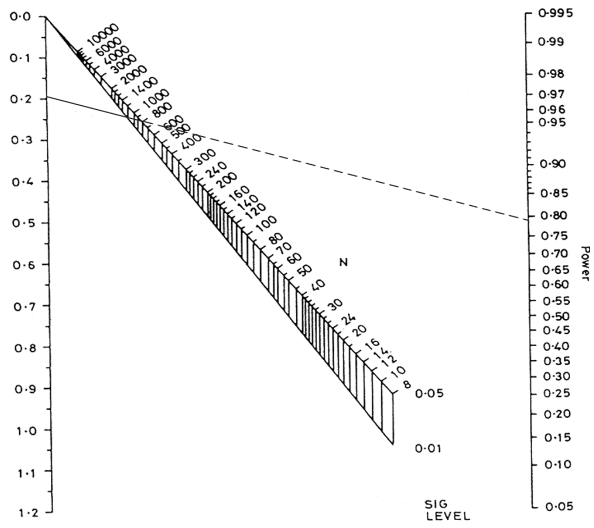
Suppose a study is testing the presence of the effect of a treatment using two groups. The study can either conclude that the two groups are substantially different or conclude that no such differences exist. What the author of the study says is given in column 1. Irrespective of what the researcher says, the fact of the issue may be either (i) the treatment is not effective – in which case no true difference is present between two groups or (ii) the treatment is effective – in which case true difference exists. The true state of affairs is plotted in row 1.
Now let us see what happens when the rows meet the columns.

	No true difference	True difference exists
Difference found in the study	**False-positive result** **Type I error** Probability of this is called α (or p-value or 'chance finding')	**True-positive result** The ability to obtain this is called the study **power** = 1 - β
No difference found in the study	**True-negative result**	**False-negative result** **Type II error** Probability of this is called β

Generally speaking, attempts to lower the type I error often increases the type 2 error and vice versa.

SPMM Memory Corner

Probability of Type 1 error = α or p value:
Probability of Type II error = β
Traditionally α = 5%; β = 20%; so power = 80%



Power:

- It is defined as the ability of a study to detect a difference between two groups if such difference truly exists.
- Typically beta is set at 0.2 and power at 80% (0.8).
- Power depends on various aspects especially sample size (n), mean effect difference that could be clinically useful (effect size), variability of observations (lower the variability, higher the power) and acceptable level of statistical significance (p).
- The term power calculation is a misnomer. In fact it is sample size calculation using set values (chosen values) of α (p-value – generally 0.05) , power (1-β; generally 80%), absolute effect size which the researcher considers as clinically relevant, and an estimated measure of variance (e.g. SD estimated from a pilot study or from literature search).
- Ideally small pilots must be run to find the variance; else they are determined from previously published works in similar clinical samples.
- The standardised difference is an expression of effect size given by (target difference in means / SD of observations). Using the standardised difference and power values, Altman constructed a nomogram that can be used to calculate sample size.
- **Methods to increase power:**

Use a larger significance level (higher p value). But this may increase type 1 error.

Use a larger sample size. This is more expensive but will reduce variance

Use larger effect sizes (consider only larger deviations from null hypothesis to be significant). This may not be desirable if a small difference can produce huge clinical impact.

Reduce variability, e.g. by making more precise measurements or matching subjects.

An one-sided test is more powerful than two sided test: check if it is possible to make strong (supported) assumptions

Use the most powerful test that the appropriate assumptions will allow. Some tests (esp.parametric) are more powerful and robust than others.

Confidence intervals

When we conduct a study, we examine a "sample" which we think is representative of the "population" from which the sample is taken.

Generalisation - We can generalise the results of our study to the "population" only if the "sample" is reasonably representative of the "population". Unfortunately, this representativeness can never be perfect. The data from our "sample" can only be approximations of the true values of the measure seen in the "population". From this, it is logical to assume that – the closer the approximations are to the "population" – the better the representativeness of the "sample".

Confidence intervals: To see how close the approximation of a measure in a sample is to the population, researchers often make use of confidence intervals.

Consider the following example: We take a "sample" of 10 students from a batch of 1000 students ("population"). If we say the mean height of boys in our "sample" was 170 cm and the 95% CI was 160 –180 cm, we mean that we are 95% confident that the mean height in the population lies between 170cm and 180 cm. In other words - if you look at the mean height of boys in 100 independent "samples" taken one after another from the population, 95/100 times, the mean value would fall between 160 and 180cm. From this example, it can be assumed that the closer the interval, the better the representativeness i.e. if the 95% confidence interval was 165 – 175 cm, our sample mean would be more representative of the population mean.

The confidence interval can also be calculated for point estimates of effect sizes, relative risks, odds ratio, NNT, etc.

<u>Interpretation of confidence intervals:</u>

In the general population, the mean paternal age at birth is 28.There is some suggestion that older fathers have a higher risk of having children with psychosis. Suppose we are interested in determining the mean paternal age of schizophrenia patients. 50 schizophrenia patients are recruited. It is found that the mean paternal age in schizophrenia group is 29.50 years. Using a standard formula, the 95% confidence interval for the mean is estimated to be 25.10 to 32.50.

In this example, 25.10 and 32.50 are the lower and upper limits of confidence; the set of values between these two constitute the 95% confidence interval.

When interpreting a confidence interval, we must look for the following:

- ***The degree of confidence***: Most often, as in the above example, a 95% degree of confidence is reported. This derives from the complement of conventional p-value or probability of type 1 error (alpha) which is 5%. One can set a more stringent p-value, and thus, a higher degree of confidence (99%). For a higher degree of confidence, a wider interval will be seen.
- ***The width of the interval***: A wide interval at a fixed degree of confidence indicates that the estimate is not precise; a narrow interval indicates a very precise estimate. The width of the confidence interval depends on the size of the standard error (variability), which in turn depends on the sample size. Therefore, small studies give wider confidence intervals than larger studies with less variability in the collected data.
- ***The upper and lower limits***: In the above example, we are 95% certain that the true mean paternal age for schizophrenia patients lies between 25.10 and 32.50 years. As these limits include the general population value of 28, there is no evidence from this study that paternal age at birth for schizophrenia patients is higher.

- ***Capturing the value of no difference***: If the 95% CI cross '0' point for the difference between means then the results is statistically not significant. If it crosses value 1 for ratio measures such as OR, etc. then again it is not significant. If it crosses infinity for inverse ratios such as NNT, then it is not significant. This is called *value of no difference* (i.e. 0 for means; 1 for ratios; ∞ for NNTs)

How can you reduce the width of the confidence interval?

Choose a smaller degree of confidence level (e.g. 90% instead of 95% confidence)
Reduce the standard deviation (e.g. by using stratification approach)

Take larger samples to increase N

Measure	Value of no difference in confidence intervals
OR odds ratio	1
Relative risk	1
Absolute risk reduction ARR	0
Relative risk reduction	0
NNT	∞

SPMM Memory Corner

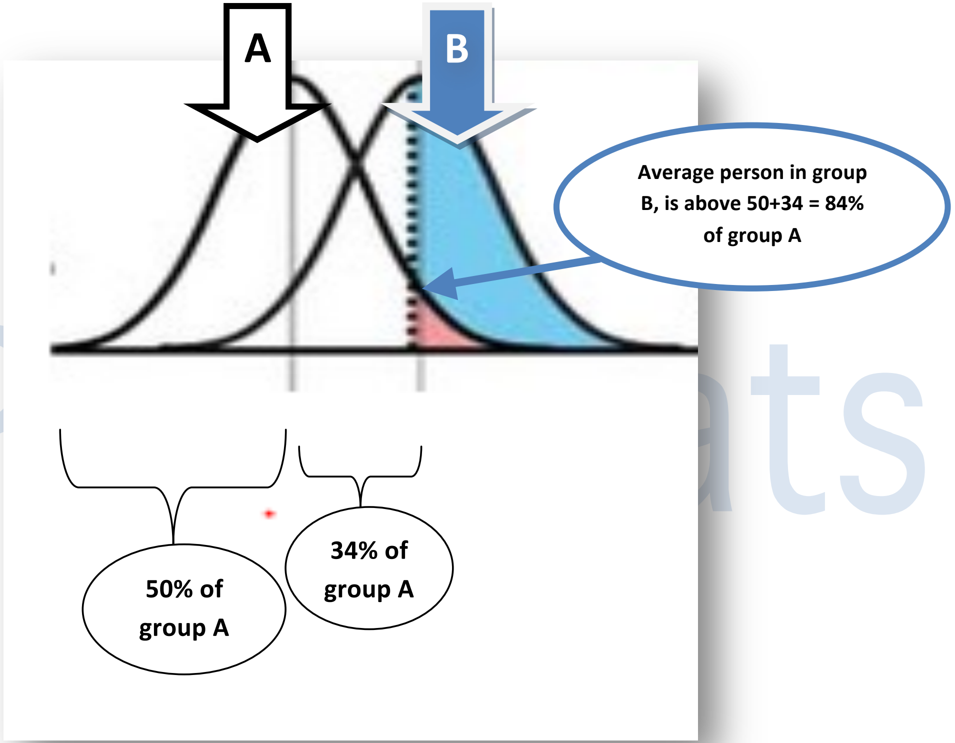
Confidence intervals inform us about
1. Degree of confidence on our sample 2. Precision of a result
3. Clinical significance (Magnitude of an effect) 4. Statistical significance

Effect Size

- Effect size (ES) can be simply defined as the difference in outcomes between intervention and controls divided by standard deviation. It is a measure of the difference in the point estimates.
- In fact, ES refers to a group of indices differing in the mode of measurement of the magnitude of treatment effect.
- Unlike p-values, these indices are independent of sample size.
- 'ES measures are the common currency of meta-analysis studies that summarise the findings from a specific area of research' (web.uccs.edu/lbecker/Psy590/es.htm). More on this can be found in meta-analysis chapter.
- These are especially useful in the meta-analysis as individual studies often report the outcome in terms of different scales – using a common measure such as effect size will help consolidating the findings.
- ES can be measured using **Cohen's d.** This is the standardised difference between two means. Cohen's *d* is the difference between the means, $M_1 - M_2$, divided by standard deviation, s, of either group.
- An arbitrary grading of effect sizes is as follows; "small, $d = .2$," "medium, $d = .5$," and "large, $d = .8$".
- **Interpreting effect size**: Assuming that 'control' and 'experimental' group values are normally distributed with same standard deviations, an effect size can be interpreted just like 'Z-scores' of a Standard Normal Distribution. For example, an effect size of 1 means that the score of the average person in the experimental group is 1 standard deviation above the average person in the control group. In other words, the average person's score in the experimental group will exceed the scores of 84% of controls. This number 84% is derived from 2 other numbers – 50% and 34% (see the attached figure).

McGraw and Wong (1992) have suggested a **'Common Language Effect Size' (CLES)** statistic, which is readily understood by non-statisticians. This is the probability that a score sampled at random from the experimental group will be greater than a score sampled from controls. e.g., the heights of young adult males and females differ with an effect size of about 2, which translates to a CLES of 0.92. In other words 'in 92 out of 100 random blind dates among young adults, the male will be taller than the female.' (Excerpt retrieved from http://www.leeds.ac.uk/educol/documents/00002182.htm)

Effect size	Percentage of controls that would be below the average person in interventional group	Equivalent CLES
0	50%	0.50
0.1	54	0.53
0.5	69	0.64
1	84	0.76
2	98	0.92
3	99.9	0.98


A
B
Average person in group B, is above 50+34 = 84% of group A
50% of group A
34% of group A

Multiple testing

- Using statistical tests multiple times with a given dataset is prone to errors.
- As by definition if p-value is set at 0.05, when one does 20 calculations with a data, he/she will get one of those positive just by chance.
- This occurs especially when one attempts subgroup analysis and testing secondary research hypotheses.
- **Bonferroni correction**: To correct for this multiple testing Bonferroni method can be used – it is rather rigorous and strict and may lead to many false negatives.
- According to this, the significance level for multiple tested data is altered as (normal significance level/number of statistical analyses carried out).
- Hence if p = 0.05 normally, and if 10 sub-analyses are attempted, then p becomes 0.005. This may be an overcorrection especially when outcomes are associated with one another. Bonferroni treats each outcome as an individual event. Alternatives to Bonferroni are highlighted below.
- **Family-wise Error:** Family-wise error (FWE) represents the probability that any one of a set of comparisons or significance tests is a Type I error.
- **Scheffe test, Tukey's honestly significant difference test & Dunnet test** are also useful.
- **False Discovery Rate (FDR**) is a new approach to the multiple comparisons problem. Instead of controlling the chance of any false positives at all as Bonferroni does, FDR controls the expected proportion of false positives.

Reliability & Validity

When developing measurement scales, we are concerned about two important properties. Can we use this scale to measure the actual phenomenon we want to measure? Can this scale provide consistent results when it is used? A highly valid scale will measure what it is supposed to measure – the truth. A highly reliable scale will provide consistent results.

Reliability refers to the replicable nature of research studies / tools. Note that high reliability does not guarantee scientific validity but guarantees consistency.

- Reliability can be assessed by **test-retest correlation** by administering an instrument twice to the same population. The time difference between test and retest must be long enough to avoid practice effect, but short enough so the underlying state (e.g. depression) does not change very much: 2 to 14 days range is often used in psychiatry.
- **Cronbach's alpha** measures the **internal consistency** of a test by correlating each item with the total score and averaging the correlation coefficients. It can take values between negative infinity and 1 as a maximum, but only positive values make sense. Arbitrary cut-off of 0.70 is used commonly to call the evaluated test to be internally consistent.
- The **split-half reliability** refers to splitting a scale into two parts and examining the correlation.
- **Interrater reliability** is measured using two or more raters rating the same population using the same scale.
- **The intraclass correlation coefficient** is used for continuous variables; it is nothing but the proportion of total variance of the measurement that reflects true between subject variability. It ranges between 0 (unreliable) and 1 (perfect reliability). ICC can be measured for either relative agreement or absolute agreement; the relative ICC is always higher than the absolute ICC. ICC of 0.6 is considered fair while 0.8 is very good and 0.9 as excellent, arbitrarily. ANOVA intraclass coefficient is used for quantitative data with more than 2 raters/groups.
- For nominal data that has more than two categories, a **kappa** or weighted kappa can be used. (More details are given below)

Validity of an instrument is the extent to which an instrument measures what it proposes to measure.

- **Face validity** refers to a subjective measure of deciding whether the test measures the construct of interest on its face value. e.g., Hamilton depression scale clearly has a face value for measuring depression; but not for measuring obsessions.

- **Construct validity** measures whether a test really measures the (theoretical) construct of interest or something else. One way of classifying the construct validity is considering unified construct validity. Here construct validity is taken to consist of both content validity and criterion validity (referred as unified construct validity).
    - **Content validity** refers to whether the contents i.e. each individual subscales, items or elements of the test are in line with the general objectives or specifications the test was originally designed to measure. It looks for a good coverage of all domains thought to be related to the measured condition. This often cannot be statistically tested, but experts are called for comments on this aspect of validity.
    - **Criterion validity** refers to the performance against an external criterion such as another instrument (concurrent) or future diagnostic possibility (predictive).
        - **Concurrent validity** refers to the ability of a test to distinguish between subjects who differ concurrently in other measures (using other instruments). e.g., those who score high on a scale of insomnia may score high on the scale of fatigue ratings too.
        - **Predictive validity** refers to the ability of a test to predict future group differences, according to current group differences in score. e.g., high aggression score in childhood and high criminal incidents in adult life. (On a similar note, **Incremental validity** refers to the ability of a measure to predict or explain variance ***over and above*** other measures)

Another way of considering the construct validity is by classifying it to convergent, discriminant and experimental/interventional validity:

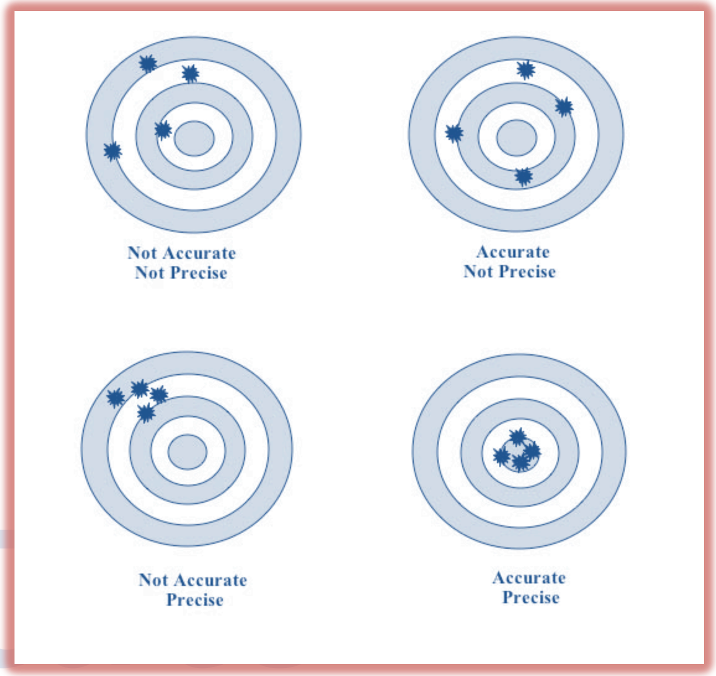
- **Convergent validity** refers to agreement between instruments that measure same construct e.g. between BDI and HAMD for depression. This agreement can be tested in contrasted groups i.e. depressed and non-depressed, both groups showing a high correlation between the two scales.
- **Discriminant validity** refers to the degree of disagreement between two scales measuring different constructs. E.g. to say that HAMD measures some construct (depression) different from that measured by Hamilton Anxiety scale (anxiety) poor correlation must be demonstrated between HAMD and HAS
- **Experimental validity:** This refers to the sensitivity to change. An instrument must show the difference in results when an intervention is carried out to modify the measured domain.

Note: ***Factorial validity*** is a form of construct validity established via factor analysis of items in a scale.

Precision and accuracy

Precision is the degree to which a calculated central value (e.g. mean) varies with repeated sampling. The narrow the variation, the precise the value is. Random errors lead to imprecision.

Factors reducing precision includes 1. Having wider the limits of the interval 2. Expecting higher confidence interval (e.g. 99.7% versus 95%). Accuracy refers to the correctness of the mean value – i.e. how close is it to the true population value. Precision is comparable to reliability while accuracy is comparable to validity. Bias in a study compromises validity / accuracy.



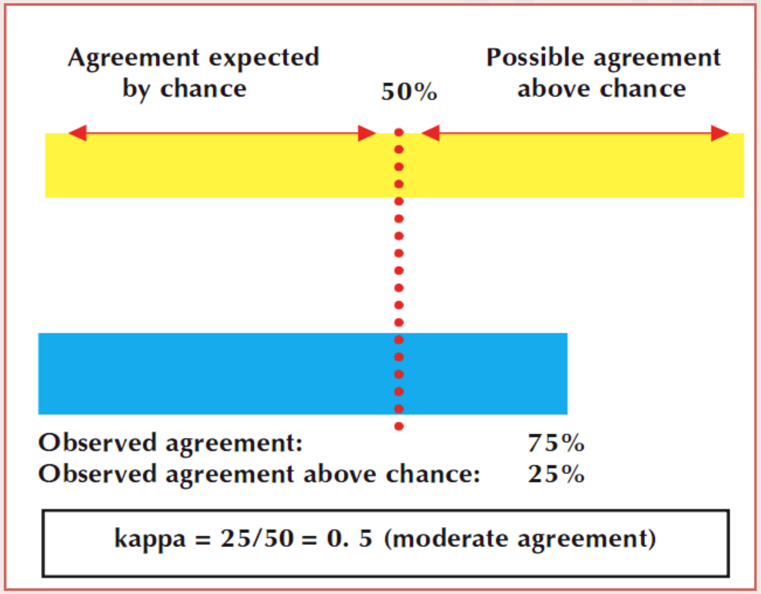
VALIDITY	QUESTION IT ANSWERS
Face	Does this scale appear to be fit for the purpose of measuring the variable of interest?
Content	Does this scale appear to include all the important domains of the measured attribute?
Criterion	Is the scale consistent with what we already know (concurrent) and what we expect (predictive)?
Convergent	Does this new scale associate with a different scale that measures a similar construct?
Discriminant	Does the new scale disagree with scales that measure unrelated constructs?

Kappa statistics

Two psychiatrists independently assess 10 consecutive patients in an inpatient unit. One of them, Dr.A diagnoses 2 patients to have hypomania. The second psychiatrist (Dr.B) is in a hurry and does not pick symptoms of hypomania in any of the ten patients. How can we characterize the level of agreement between these 2 clinicians?

The percent agreement: The percent agreement between them is 80% (as they have agreed in 8 out of 10 instances with respect to presence or absence of hypomania) even though one of them has been unacceptably careless. Hence, measuring the simple percent agreement overestimates the degree agreement and may be misleading. What we really need is how much the 2 observers actually agree beyond the level of agreement that could be expected by chance.

Beyond-chance agreement: Kappa indicates the level of agreement that could be expected beyond chance.



Kappa is calculated only for agreement on categorical variables such as presence or absence of a diagnosis. For ordinal variables, weighted kappa is used. Bland-Altman plot is used for continuous variables.

Interpreting kappa:

Kappa is affected by the prevalence of the outcome studied – i.e. higher the proportion of positive (or negative) assessments, higher the kappa.
Statistical significance cannot be tested directly from kappa measures: we can use kappa as a guide to actual degree of agreement using the following table:

Kappa value	Degree of agreement
0	None
0–0.2	Slight
0.2–0.4	Fair
0.4–0.6	Moderate
0.6–0.8	Substantial
0.8–1.0	Almost perfect

Calculating Kappa:

$$\text{Kappa } \kappa = \frac{\text{Observed Agreement beyond chance /}}{\text{Maximum agreement beyond chance}}$$ OR

$$\text{Kappa } \kappa = \frac{(\text{Observed agreement} - \text{ agreement expected by chance})}{/\ (100\% - \text{ agreement expected by chance})}$$

NOTE: Chance agreement is not always 50%; rather, it varies from one clinical situation to another.

To calculate kappa we need 2 numerical values: the percentage of patients that the 2 assessors correctly classified and the expected agreement by chance. Both can be determined by constructing a 2 × 2 table:

Suppose Drs. A and B are attempting to diagnose hypomania by clinically interviewing 100 patients independently.

OBSERVED DIAGNOSIS		Dr B		total
		Hypomania	No Hypomania	
DrA	Hypomania	9	3	12
	No Hypomania	16	72	88
Total		25	75	100

Observed agreement = 72+9/100 = 81%

Suppose if Dr. B is making a diagnosis of hypomania purely by chance, without any expertise. From the above observations, it seems that he is classifying 25% of his patients to have hypomania. If he depends on some dirty trick and classifies 25% of any sample to have hypomania, then we would expect him to 'diagnose' hypomania in 3 out of 12 patients labelled as hypomanic (25% of 12 is 3) by Dr. A and 22 out of 88 patients dismissed as not hypomanic (25% of 88 is 22) by Dr. A. We can expect the following:

EXPECTED DIAGNOSIS		Dr. B (The 25:75 Doctor!)		total
		Hypomania	No Hypomania	
Dr.A	Hypomania	$\frac{25}{100}\times 12$	?	12
	No Hypomania	$\frac{25}{100}\times 88$	?	88
Total		25	75	100

EXPECTED DIAGNOSIS		Dr. B (The 25:75 Doctor!)		total
		Hypomania	No Hypomania	
Dr. A	Hypomania	3	9	12
	No Hypomania	22	66	88
total		25	75	100

Expected agreement = 3+66/100 = 69%

$$\textbf{Kappa } \kappa = \frac{(81\% - 69\%)}{(100\% - 69\%)} = \frac{12\%}{31\%} = \mathbf{0.39}$$

The kappa coefficient is dependent on the prevalence of the measured condition. For rare disorders such as Capgras syndrome, good agreement can be achieved between raters, even if a strong disagreement is seen in a few cases. But for common disorders such as depression, kappa will be low. Thus, one must look at the actual percentage agreement instead of relying solely on kappa values.

Kappa is used for ***nominal*** variables (agreement of grouping categories, saying yes or no rather than on a particular value). Disadvantages are that one cannot test statistical significance directly from the kappa values.

- **Phi** is another measure that it is not as popular as kappa yet. To calculate phi all cells are utilized; statistical significance testing is possible; a small sample size can be used.
- **Weighted kappa** is used for ordinal variables.
- **Bland-Altman plot** is used for continuous measures where as pairs of score differences are plotted against the mean.

Choosing Statistical Test

There are few questions one must ask when choosing a statistical test. The most important question is

'What is being tested?' In other words 'are we comparing different samples for a point estimate such as mean, median or frequency counts (proportion) – Category 1 - or are we trying to demonstrate association and relationships? ' - Category 2

A simple clue is – studies looking for causal association will mostly fall under category 2.

For category 1 (most common in MRCPsych exams) use the following structured approach: Ask

1. What is the nature of the point estimate one is interested in?
    a. Simply - mean or proportions?
2. How many groups are being compared?
3. How many observations of similar nature are made in each subject?
    a. Simply – paired or unpaired (independent) observations?
4. Can we assume parametric distribution – is there anything that precludes this?
    a. Simply – is the data not continuous quantitative?

Statistical tests: MEANS (parametric)			
Means	One	n/a	**One sample t test**
	Two	Unpaired	**Two sample t test**
		Paired	**Paired t test**
	>2 groups	Unpaired	**One way ANOVA**
		Paired	**MANOVA, repeated measures ANOVA**

Statistical tests: Medians (nonparametric)

Medians	One	n/a											**Sign test**
	Two	Unpaired											**Mann-Whitney U test**
		Paired											**Wilcoxon rank sum test**
	> 2 groups	Unpaired											**Kruskal-Wallis test**
		Paired											**Friedman test**

Statistical tests: PROPORTIONS (categorical)

Proportions	Two	Unpaired																		**Chi Square test**
		Paired																		**McNemar test**
	>2	Unpaired																		**Log-linear or logistic regression**

Parametric tests can be used if at least one variable is quantitative and normally distributed (note that most biological variables are normally distributed; even if not they could be transformed into normally distributed values (see below)). Unless explicitly stated as distribution not known, assume all biological variables are normally distributed for the sake of choosing a statistical test in EMIs/BOFs.) When comparing groups, t-tests and ANOVAs are commonly used (more details in 'Comparing Means' chapter).

Non-parametric tests:

- ♣ These are used when (a) both dependent and independent variables are qualitative (nominal or ordinal) or (b) when the variables are quantitative but not normally distributed.
- ♣ In nonparametric tests **ranks** (not just medians) are compared – it is possible to have groups with similar means but different ranks.
- ♣ The **sign test** is a simple non-parametric test that compares the median of a sample to the median of the population. It is the non-parametric equivalent of one sample t test.
- ♣ **Wilcoxon rank sum test** is used for comparing two paired observations in nonparametric fashion; thus it is equivalent to paired t.
- ♣ **Mann-Whitney** U test is used for independent observations in two groups.
- ♣ **Kruskal-Wallis** is more or less non-parametric one-way ANOVA equivalent used for 3 or more groups.

Transformation of data:

Parametric statistical tests offer more robust analysis than non-parametric tests. Hence, one may convert data into a form more suitable for parametric analysis using transformation methods. The most commonly used method is **log transformation**. For a right skew (positive) this yields a normal distribution (called ***lognormal curve***). Such transformation can also linearize a nonlinear correlation with the upward shift. **Square root transformation** has normalizing and linearizing properties; in addition it can stabilize variance. It is used for normalizing Poisson distributions. Other types include reciprocal transformation**, square transformation,** and logit transformation. The **reciprocal transformation** is often used for analysis of survival rates. The **logit transformation** is often used for distribution of proportions. It linearizes a sigmoid curve. Note that confidence intervals may be difficult to interpret after transformation. Geometric mean values may get distorted in the process to some extent.

Chi-square χ^2 test is commonly used non-parametric test, especially for comparing frequency counts or proportions.

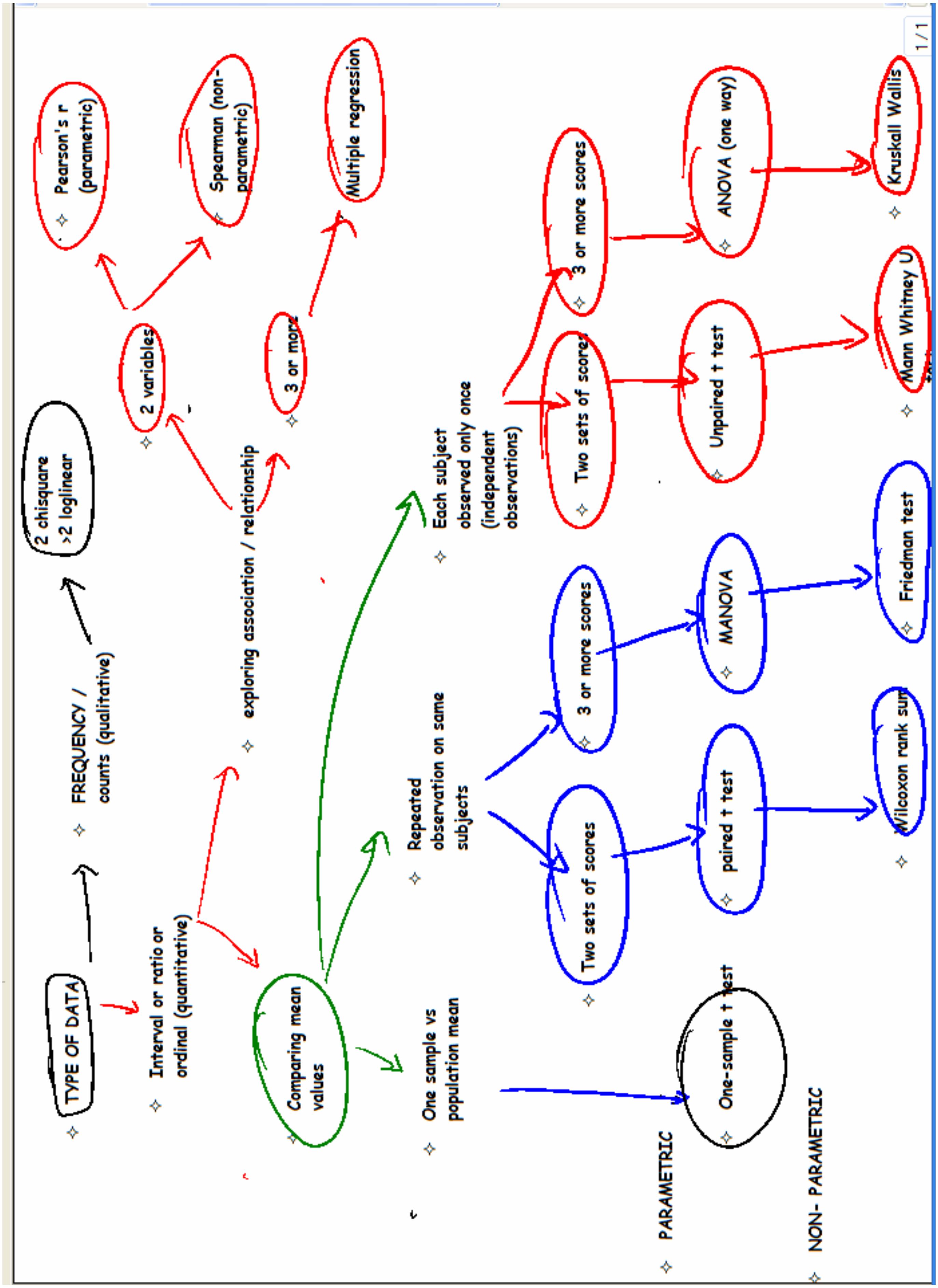
- A **contingency table** is constructed when using chi-square test. A table whose contents are in frequencies is called contingency table.
- When one variable is studied – there are two outcomes possible. If there are 2 groups, then we can construct a **2X2 table.**
- The rows (and columns) must be mutually exclusive i.e. no individual must be represented in more than one cell.
- Observed frequencies are the real outcomes observed at the conclusion of a study. Expected frequencies are the outcomes predicted as if the null hypothesis was true.
- The ratio of **observed to expected** frequencies in the cells of the table is called chi-square.
- A **Fisher exact test** is used in place of chi-square if the expected cell frequencies in more than 20% of cells fall less than 5. In other words, for a 2X2 table, even if one cell (which constitutes 25%) has expected frequency < 5, then corrections must be used.
- **Yates correction** can be used if the total sample was less than 100 or any cell value was less than 10.
- χ^2 must always be applied to *original frequency* counts and not after converting these to percentages as this will reduce the cell values, and incorrect results are obtained.
- **McNemar's** test is χ^2 test for paired data.
- One form of chi-square test wherein the influence of two dichotomous (binary) categorical variables on one dependent variable is tested is called **Mantel-Haenszel test.**
- Another special form of chi-square test where log values of cell frequencies are employed is called **log-linear analysis.** This is often used when more than 2 groups or variables are studied.

Category 2:

1. Decide how may dependent variables (usually 1)
2. Decide how many independent variables
3. Check if they are categorical or continuous

Statistical tests: Testing association

Number of dependent variables	Number of independent variables	Parameter tested	Nature of data										Test
One	1	Correlation	Continuous, parametric										**Pearson's**
			Non parametric										**Spearman's, Kendall's tau**
		Regression	Continuous										**Simple linear regression**
			Categorical										**Simple logistic regression**
One	>1	Regression	Continuous										**Multiple linear regression / ANCOVA**
			Categorical										**Multiple logistic regression**
Two or more	>1	Regression	Categorical										**MANOVA**
			Continuous										MANCOVA/or multivariate multiple regression


TYPE OF DATA
FREQUENCY / counts (qualitative)
2 chisquare
>2 loglinear
Interval or ratio or ordinal (quantitative)
exploring association / relationship
2 variables
Pearson's r (parametric)
Spearman (non-parametric)
3 or more
Multiple regression
Comparing mean values
One sample vs population mean
Repeated observation on same subjects
Each subject observed only once (independent observations)
Two sets of scores
3 or more scores
Two sets of scores
3 or more scores
PARAMETRIC
One-sample t test
paired t test
MANOVA
Unpaired t test
ANOVA (one way)
NON- PARAMETRIC
Wilcoxon rank sum
Friedman test
Mann Whitney U
Kruskall Wallis
1 / 1

Comparing Means

Statistical tests are often employed to compare two or more populations or two or more experimental conditions. usually this is achieved by comparing the means of the different groups.

1. **One sample t-test:**
   - Used to compare **mean of a single sample** with a known population mean to check if the sample is derived from the described population.
   - It needs two values – an observed sample mean, and hypothesized / elsewhere derived population mean.
   - The variable must be normally distributed in the population.
   - Sample size must be adequate enough to prevent extreme skewness.

2. **Two sample t test (Student's t)**:
   - Used to compare means of two samples.
   - It needs two variables – one independent variable, which can be nominal, and a dependent variable, which is quantitative.
   - **Unpaired** t test compares independent samples where observation is made only once in each individual in a sample.
   - In a **paired** t test, sample means are compared in the same group i.e. pretest vs. posttest. Here each subject is observed twice, say, before and after an intervention and the scores are compared for statistical significance.
   - The t-test assumes **equal variance** among the two groups of given data. To test, this **Levene's test** can be used.

3. **ANOVA**
   - Is used to compare means of multiple groups; it is a fairly robust test.
   - It is based on variance comparison.
   - If just one independent variable is compared across more than 2 groups, this is called **one-way ANOVA**.
   - In **two-way ANOVA,** 2 independent variables are analyzed in multiple groups. E.g. in a study analyzing the effect of long DUP on treatment response in three groups – adolescent, middle age and late onset schizophrenia, two independent variables such as long vs. short DUP and male vs. female sex can be considered.
   - **Repeated measures ANOVA** is used when the same measure is used on multiple occasions – i.e. paired observations.
   - ANOVA is calculated using the ratio of variation between groups to the variation within groups. This is called **F statistics.**

- A disadvantage with ANOVA is that it can only tell us if a significant difference exists among groups; it does not point out from where the difference comes from. Hence, **post hoc significance tests** such as multiple t-tests are used.
- Assumptions for using ANOVA include
    - Normal distribution of observations in groups
    - Equal variance among the tested groups
    - Independent observations (not paired – except for repeated measures test)

Number of groups	**Number of independent variables**	**Between subjects or within-subjects?**	**Recommended test**
>2	1	Between	One way ANOVA
>2	1	Within	Repeated measure ANOVA
>2	>1	Between	Factorial ANOVA
>2	>1	Within	MANOVA

Degrees of freedom

The concept of degrees of freedom is fundamental for estimating statistics of populations from samples (inferential statistics). "Degrees of freedom" is commonly abbreviated to df. In short, df can be considered as a mathematical required when we *calculate an estimate* one statistic from an *estimate* of another.

Let us take an example of calculating standard deviations. When we calculate SD of a sample, this depends on randomly distributed individual observations. There is no restriction in this calculation, so we use the actual sample size as denominator.

$$\sigma = \sqrt{\frac{\sum (x-\mu)^2}{n}}$$

[x is a value from the sample, m is the mean of all x, n is the number of x in the sample]

But if we calculate a population's SD from the estimated SD of a sample (as given below in the formula), our calculation is restricted by a factor of one – the factor being the sample mean (x bar) that has already been estimated. Hence, we use n-1 as the degree of freedom.

$$s = \sqrt{\frac{\sum (x-\bar{x})^2}{n-1}}$$

[x is an observation from the sample, x-bar is the sample mean, n is the sample size]

When this principle is applied to regression or ANOVA, we lose one degree of freedom for each parameter estimate carried out before estimating the (residual) standard deviation.

The restriction principle behind degrees of freedom is comparable to the concept of contingencies. We have four numbers (x, y, z, and k) that must add up to a total of n; we can freely choose the first three numbers at random, but the fourth choice is constrained or restricted so that it makes the total equal we require- thus the degree of freedom becomes three in this instance.

A 2x2 table such a the one used in computing chi-square statistic generally has 4 degrees of freedom - each of the 4 cells can contain any number. If row and column marginal totals are specified, there is only 1 degree of freedom. So for Chi-Square distribution where proportions are estimated, the degree of freedom is (number of rows -1) X (number of columns-1). This is because the total proportion is always expected to make up to 100%; in a 2X2 chi-square, only one of the two rows has the freedom to be of any value, the other can only be (100-the previous value).

We sometimes need to determine the degrees of freedom (df) for the t test. In the t-test, the degree of freedom is the sum of the persons in both groups minus 2.

Estimate	df
Sample mean	n
Population SD estimated from sample	n-1
Two sample t test	(n1+n2)-2
Chi-square	(Rows-1)*(columns-1)
One sample t	n-1
One way ANOVA: total variance	Total N - 1
One way ANOVA: within groups	Total N – number of groups
One way ANOVA: between groups	Number of groups - 1

Regression & Correlation

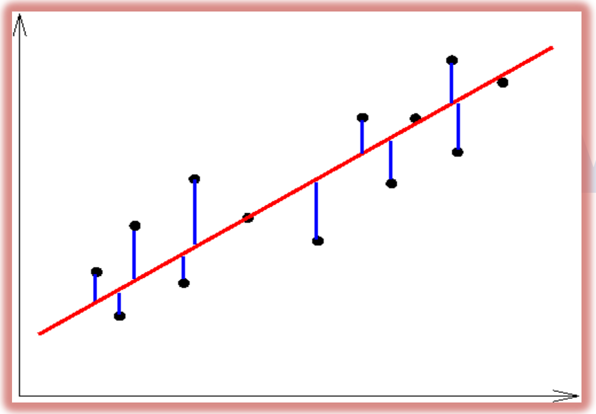
Correlation:

- Correlation refers to the degree to which two quantitative variables are related / associated.
- A **scattergram** or plot can show the relation between two variables.
- **Pearson's correlation coefficient** is commonly used measure for quantitative parametric data.
- If variable A increases in value whenever variable B also increases, then A and B are **positively correlated** (+).
- If the variable A decreases whenever variable B increases, this is a **negative correlation** (-).
- If no such relationships exist in either direction, this is called the absence of correlation (0).
- Values of correlation coefficient 'r' thus range from -1 to +1. The ***vector (sign)*** indicates the direction of the relationship. The ***magnitude*** indicates how close the relationship is to linearity.
- It has no **units** for measuring.
- Note that A and B can be interchanged, but this will not result in a change in vector or magnitude of r.
- Correlation does not suggest causation. The correlation coefficient is not valid if the linear relationship is not suspected or if the data is not independent (paired).
- **Fisher's transformation** may be used to compare two correlation coefficients for hypothesis testing.
- **Partial correlations** are correlations between two variables after adjusting for a third variable.
- **Spearman's correlation** (rho) is the non-parametric equivalent of Pearson's. It can be used to test the association between two variables if at least one is ordinal (ranked); or if sample size is small despite being continuous variables; or if non-linearity is suspected; or if non-normal distribution is noted for both variables. Spearman's rho assumes that the difference between each pair of ordinal variables is the same (i.e. the ranks are equidistant). If this is not true, then **Kendall's tau** will be a more appropriate measure of nonparametric correlation.

Regression:

- From correlation statistics, we can say if two variables are associated and if so, in which way and to what extent. But we cannot predict what value one variable will take if given a particular value of the other. In order to make such predictions, we need to construct regression statistics.
- For such predictions, we need mathematical expression (though we hate them!). Consider one such equation $\mathbf{y = a + bx}$; here 'b' is called **regression coefficient,** and 'a' is the intercept on the y-axis. This is called a **simple linear regression.**
- Such linear regression statistics can predict probable score in Y axis from a known score in X axis i.e. dependent variable can be predicted from value of independent variable –

- Such predictions are not possible with mere correlation – one requires a correlation + regression for this. But correlation is a prerequisite for having a regression equation between two variables.
- In the above equation how do we determine the value of a and b? This is usually done using a scattergram and **method of least squares**. In this method, a hypothetical straight line is so constructed that its vertical distance from various points of observations (the dots in a scattergram) is kept to a minimum. This vertical distance is called residue. The sum of the square of residues is kept to a minimum for a regression line of **good fit.**
- In **multiple linear regression,** several independent variables together predict a single dependent variable. Consider $\mathbf{y = a+b_1x_1 + b_2\,x_2+ b_3x_3+ b_4x_4 \ldots\ldots + b_nx_n}$.
- Multiple regression is a multivariate technique. For other multivariate techniques see below.
- In multiple regression, these independent variables are often called **covariates**.
- In some instances, among many covariates studied, two may be highly correlated with each other. This is called as **collinearity,** and it may disturb the regression.
- The regression coefficient is useful to examine confounders; confidence intervals can be used in expressing the correlation coefficients.



→ Least squares method

Each dot is a value of dependent variable corresponding to the observed independent variable. The vertical lines are residues. The linear line is constructed such that the sum of the square of residues is kept minimal. Note that statisticians prefer **maximum likelihood estimation** method for regression analysis.

- R^2 – square of regression coefficient - also called as coefficient of determination – is used to test the **goodness of fit** or final regression. It is the proportion of total variation in the dependent variable that can be explained by the independent variable. It measures how well actual outcome (dependent variable) and calculated dependent variable correspond with each other.
- R^2 ranges from 0 to 1. There will be many R^2 values for multiple regression, but their sum will not exceed 1. R^2 can also be calculated for Pearson's correlation coefficient 'r', in which case it is called the coefficient of variation.
- For linear regression, the dependent variable must be continuous. If the dependent variable is binary, then the **logistic regression** is used. It is popular as it can give OR, RR and hazard ratio for independent variables that affect the dependent variable.

Nature of variable	Number of independent variables	Number of dependent variables	Test
Continuous	One	One	Simple linear regression
Continuous	>1	One	Multiple linear regression
Binary	One	One	Simple logistic regression
Binary	>1	One	Multiple logistic regression

- Note that one can use **log-linear analysis** in the same way as logistic regression but there is no clear demarcation between DV and IV in log-linear models. Logistic regression allows continuous as well as categorical independent variables to be included in the regression analysis, but log-linear accommodates only categorical data.
- Variables that have dichotomous outcomes used in the logistic regression are often called **Bernoulli** random variables. E.g. a treatment-emergent side effect (yes/no), treatment success (yes/no) etc.
- If one wants to demonstrate the exponential relationship of a variable with a factor such as time, log-transformed values can be plotted against time. This is called **exponential correlation.**
- **Polynomial regression:** In some cases of non-linearity, the relationship between dependent variable y and independent variable x could be expressed as $Y = X^n$, where n may be 2,3, 4, etc. This higher order relationship is called polynomial regression.
- **1 in 10 rule:** As a rule of thumb, the number of variables studied in multiple regression models must not be greater than 10% of sample size. For logistic regression, the number of variables must not be greater than 10% of number of events.
- **Performing multiple regressions**: This can be done in various ways.
    - ***Stepwise regression***: Calculates coefficient of regression and starts with most significant to least significant independent variable and fits them in a stepwise fashion into the regression equation. But a disadvantage is sometimes the statistically significant variables may not be clinically significant.
    - ***Forward selection***: A confounding factor by definition is associated with both independent variable and dependent variable. As one may not know which is a confounding variable in some occasions (this is what one often examines with multiple regression), these are treated as covariates. While constructing multiple regression equations, if the regression coefficient of a previously added variable changes then either one of the covariates is a confounder; so these are retained in the equation irrespective of statistical significance. The latter added covariate is discarded if no change occurs in regression coefficients.
    - ***Backward elimination***: This starts with final model – the full equation and tries to discard covariates one by one according to changes that occur in correlation coefficients.

Regression equation

Y is the dependent variable – the outcome of interest

X is the independent variable – the predictor of outcome

$$y = a + bX + e$$

E is the error. It is a random variable with mean = 0; it represents the part of the variability of Y which is not explained by the relationship with X.

a and b are constants; b is the slope or regression coefficient

SPMM Memory Corner

Using the method of least squares, we can find the best linear regression equation with minimum variance of 'e'.

Multivariate analyses

A multivariable analysis is used for data with one dependent outcome variable but more than one independent variable. The multivariable analysis is a tool for determining the relative contributions of different causes to a single event or outcome. **Multivariate analysis** is used for data with more than one dependent outcome variable as well as more than one independent variable. These terms are often used interchangeably.

If both the dependent and independent variables consist of continuous data, we can use **multiple regression.**

If the dependent variable consists of dichotomous categoric data (two outcomes) use **logistic regression.**

If the dependent variable also includes a time factor (e.g., a survival curve) use the **Cox proportional hazards** model.

If the dependent variable consists of nominal categorical data (i.e., more than two outcomes), use **log-linear analysis**.

For a continuous dependent variable with categorical independent variables, use **analysis of variance (ANOVA)**

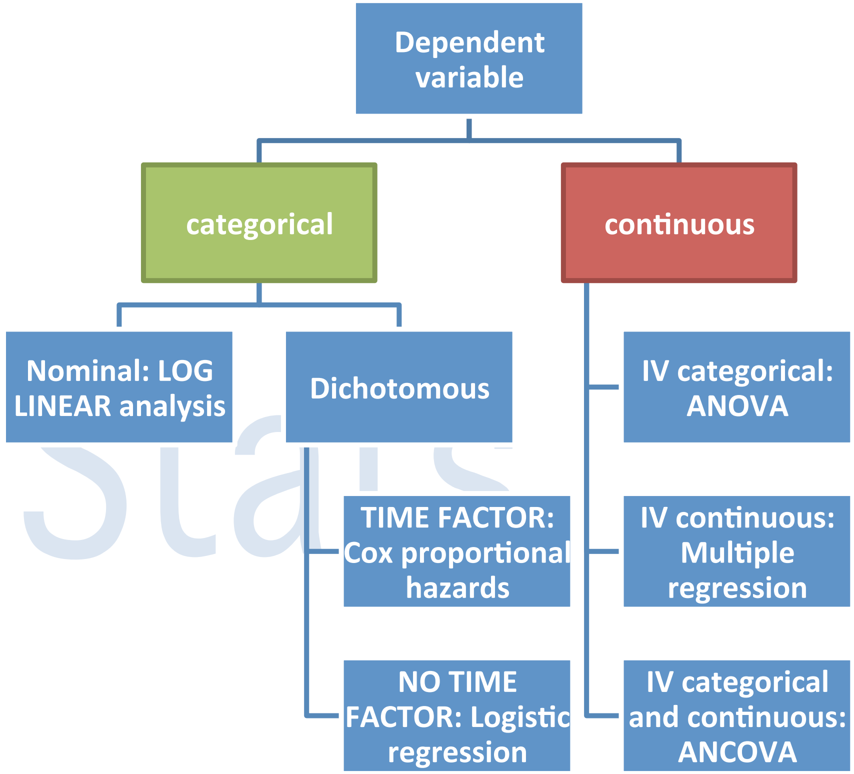
If there are both categorical and continuous independent variables, use **analysis of covariance (ANCOVA).**

Path analysis is an extension of multiple regression. It can examine situations in which there are several final dependent variables with "chains" of influence, i.e. variable A influences variable B, which in turn affects variable C, etc.

Cluster analysis, which is a multivariate tool used to organise variables into relatively homogeneous groups, or "clusters." It involves the generation of a **similarity matrix**. A cluster analysis produces a **dendrogram**

Canonical correlation, which is a multivariate tool used to explore the relationship between two sets of variables. For e.g., the relationship between (1) a list of various risk factors for schizophrenia and (2) a list of various first rank symptoms of schizophrenia. Canonical correlation involves the computation of **eigenvalues.**

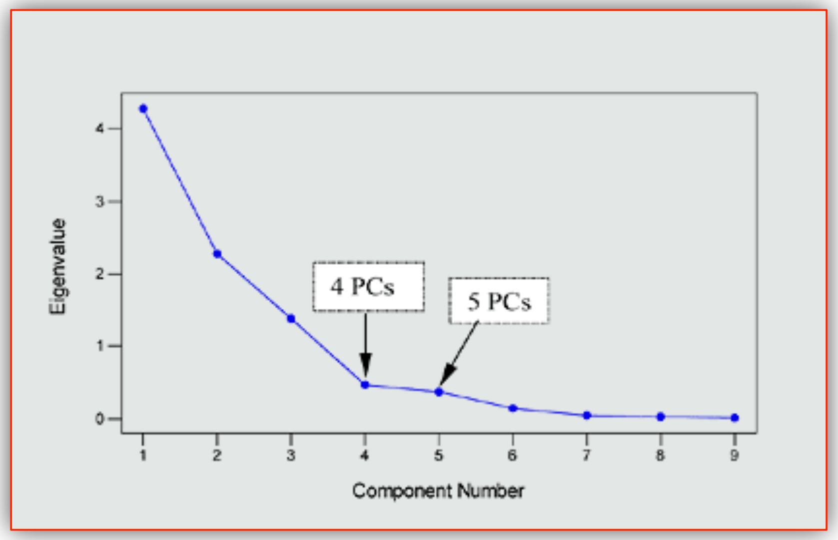
Discriminant function analysis, which is a multivariate technique used to detect which of several variables best discriminates between two or more groups. It is computationally similar to the multivariate analysis of variance **MANOVA**.



KEY USES

1. To look for interaction between independent variables.
2. To quantify associations.
3. To adjust for potential confounders in a controlled study.
4. To develop models to predict values or probabilities of certain outcomes

Factor analysis



A scree plot

Factor analysis refers to a set of statistical methods used to detect underlying patterns in the relationships among a number of observed variables. It aims to identify whether the correlations between a set of multiple observed variables can be summarised in terms of a smaller number of underlying, latent, unobserved variables called '**factors**'.

There are two main approaches: exploratory factor analysis and confirmatory factor analysis.

Exploratory factor analysis is used for the preliminary investigation of a set of multiple observed variables. It makes no a priori assumption about the composition of underlying latent variables or factors. (`Retrieved from http://ebmh.bmj.com/content/11/4/99.extract`)

The applications of exploratory factor analysis include

1. Data reduction when multiple (over25) variables have been measured, to provide a parsimonious description of the data.

2. Classification of symptoms into clinically meaningful concepts

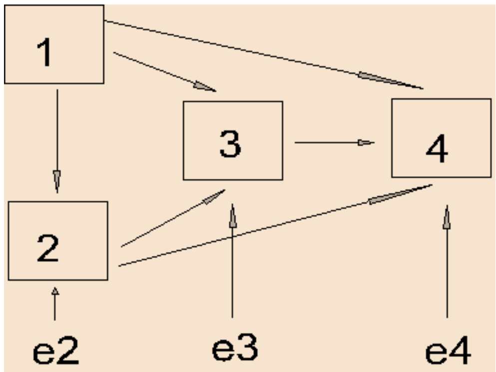
3. Definition of subscales of new measures

Confirmatory factor analysis is a method for testing whether a specified factor structure remains valid with a new dataset. Confirmatory factor analysis is primarily used for assessing the construct validity of questionnaires or tests.

Conducting a factor analysis involves a series of stages:

1. Constructing the **correlation matrix**;
2. **Extraction**: Common factor analysis and **principal components analysis** are most frequently used.
3. **Rotation** measuring the ***eigenvalues***;
4. Defining the factors to be retained: The eigenvalue is the amount of total variance explained by each factor. To determine the number of factors to be retained we can follow one of the three approaches:
   a. Include as many factors as required until an adequate preset proportion of variation is explained by them.
   b. ***Kaiser rule***: Only factors with eigenvalues greater than 1 are retained generally.
   c. ***Scree plot***: Plot the component numbers against eigenvalues. Choose the number that forms the elbow or bend before the plot levels off on the right side.
5. **Labelling** and reporting the factors: There is a general consensus that the variables with a factor loading greater than or equal to 0.40 (an essentially arbitrary but conventionally accepted cut-off) are probably making a significant contribution to that factor in contrast to those with smaller factor loadings.

Path analysis: This refers to causal modelling and prediction beyond simple regression. A set of independent variables is related to a set of dependent variables in these analyses. It can be used to examine situations in which there are various "**chains" of influence** among the variable studied. The independent (X) variables are called **exogenous variables** whereas the dependent (Y) variables are called **endogenous variables**. A path relationship is displayed using arrows that display presumed causal relations. A single-headed arrow flows from a putative cause to the effect. A double-headed, curved arrow indicates mere correlation but no predictive, causal links. A ***path coefficient*** indicates the direct effect of a variable assumed to be a cause on another variable assumed to be an effect.Path coefficients are written with two subscripts. The path from 1 to 2 is written p_{21}, the path to 2 from 1. Note that the effect is listed first. A path analysis in which the causal flow is unidirectional (no loops or reciprocal causes) is called *recursive*.



Points to note:

1. All possible paths from earlier to later variables are included in this particular graph (1 to 2, 3, & 4; 2 to 3, 4, 3 to 4). There are no backwards paths (e.g., 4 to 1).
2. The only exogenous variable is 1 (it has no arrows pointing to it).
3. There are 3 endogenous variables here (2,3,4). Each endogenous variable is explained by 1 or more variables in the model, plus an error term (e2 - e4). An endogenous variable can be a cause of another endogenous variable, but not for an exogenous variable.

(from MT Brannick; http://faculty.cas.usf.edu/mbrannick/regression/Pathan.html)

Stratification

Stratification is often used to control for analyse the effect of confounder variables. A **stratum** is a sub-group within a sample often defined by the presence or absence of a variable of interest.

While controlling for confounders, we can classify subjects, according to presence or absence of a known confounder in both study groups. Later one can carry out analysis in each sub-group or strata to find the degree of association between presumed cause and effect.

This approach will produce many sub-tables instead of one 2X2 table at the end of a case-control or cohort study. Later, we can produce a single overall estimate using various methods for obtaining summary risk values (**Adjustment**)

Table 1 gives the summary of a hypothetical cohort study that followed up two groups – one exposed to antidepressants and the other unexposed. The study concluded that the relative risk of deliberate self-harm in those taking antidepressants was 2.52.

TABLE 1	DSH	No DSH	Total	Incidence
Antidepressant exposed	45	305	350	12.8
No exposure to antidepressants	17	318	335	5.07
			685	RR = 2.52

A critique pointed out that age is a confounding factor. He divided the sample and produced the **table 2.** He showed that different age groups (strata) had different relative risks.

TABLE 2	Antidepressant exposed			No exposure to antidepressants			Total in sub-table T^s	RELATIVE RISK
	DSH D_E	Total N_E	Incidence of DSH	DSH D_{NE}	Total N_{NE}	Incidence of DSH		
25- 34	15	60	25	5	75	6.66	135	3.75
35 -44	12	90	13.3	6	80	7.5	170	1.77
45 - 54	10	90	11.1	4	75	5.33	165	2.08
55 -64	5	70	7.1	1	60	1.66	130	4.27
65-74	3	40	7.5	1	45	2.22	85	3.38
ALL AGES	45	350	12.8	17	335	5.07	--	2.52 (crude total RR)

In such cases, a crude summary may not reflect the actual risk. It is best if one applies a method of weighting for each stratum and then produce a summary score. One such method is the **Mantel-Haenszel procedure**. Table 3 shows how the adjusted RR is calculated.

TABLE 3	Total in sub-table	$D_E \times N_{NE} / T^s$	$D_{NE} \times N_E / T^s$	Mantel-Haenszel RR
25- 34	135	8.33	2.22	
35 -44	170	5.64	3.17	
45 - 54	165	4.54	2.18	
55 -64	130	2.30	0.53	
65-74	85	1.58	0.47	
SUMMATION	--	22.39	8.57	2.61 (adjusted RR)

Mantel-Haenszel can also help to find if a variable is just an **effect modifier** or a real confounder. See table 4 for further explanation.

Type of third variable	Sub-groups vs. total risk	Adjusted vs. non-adjusted risk
Neither an effect modifier nor a confounder	Sub-strata risk does not differ from crude total risk	Summary risk does not differ much from crude total risk
Effect modifier and confounder	Sub-strata risk differs from crude total risk	Summary risk differs from crude total risk
Effect modifier but not a confounder	Sub-strata risk differs from crude total risk	Summary risk does not differ much from crude total risk
Confounder but not an effect modifier	Sub-strata risk does not differ from crude total risk	Summary risk differs from crude total risk

Standardisation: It is another method used in large data sets for public health statistics to produce adjusted rates. Age is the most often standardised variable. For the purposes of standardisation, a hypothetical 'world standard population' produced by WHO is often used. There are 2 methods – direct and indirect standardisation.

- **Direct standardisation**:
  - Stratum specific rates from study sample are applied to the standard population
  - Summary score is produced from this data
- **Indirect standardisation**:
  - Stratum specific rates from the standard population are applied to the study sample; this will give 'expected rates'. Expected rates are divided by the observed rates to arrive at standardised rates e.g. Standardised Mortality Ratio (SMR)

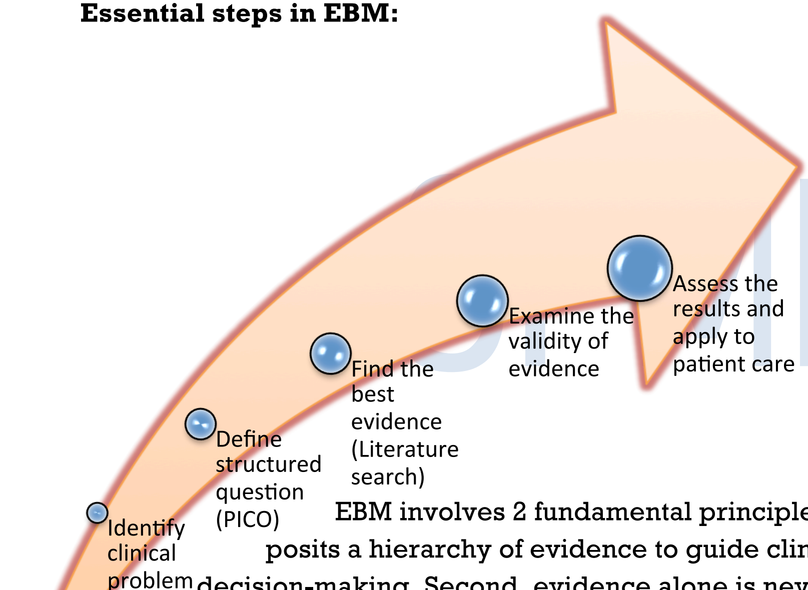
PAL Memory Corner

- Stratification is useful only for known confounders
- Adjustment can be applied to Odds ratio as well as RR
- Multivariate techniques such as regression can also be used for analysing confounders.

EBM Principles

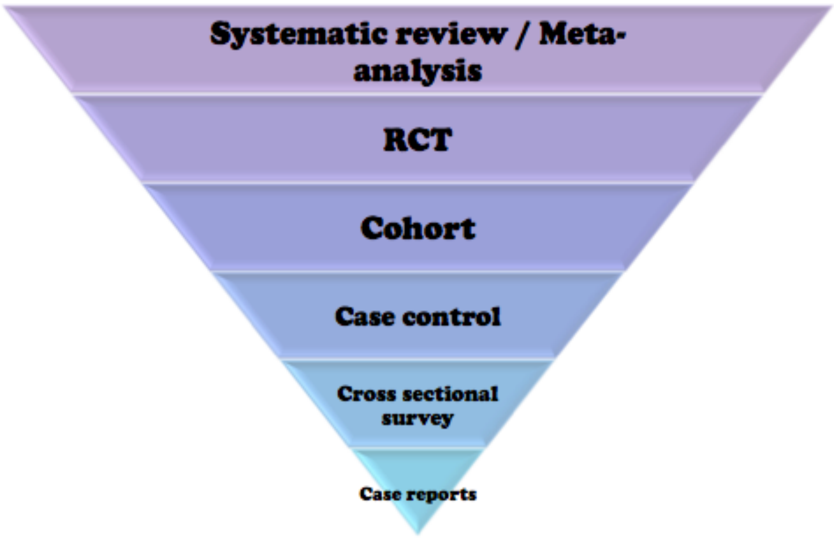
Evidence-based medicine is an approach to solving everyday clinical problems. Traditionally medical practice placed a higher value on clinical experience and pathophysiologic rationale in clinical management. But EBM stresses the importance of examination of evidence from clinical research, introducing a formal set of rules to the results of clinical research and places a lower value on authority than the traditional medical paradigm.

Essential steps in EBM:



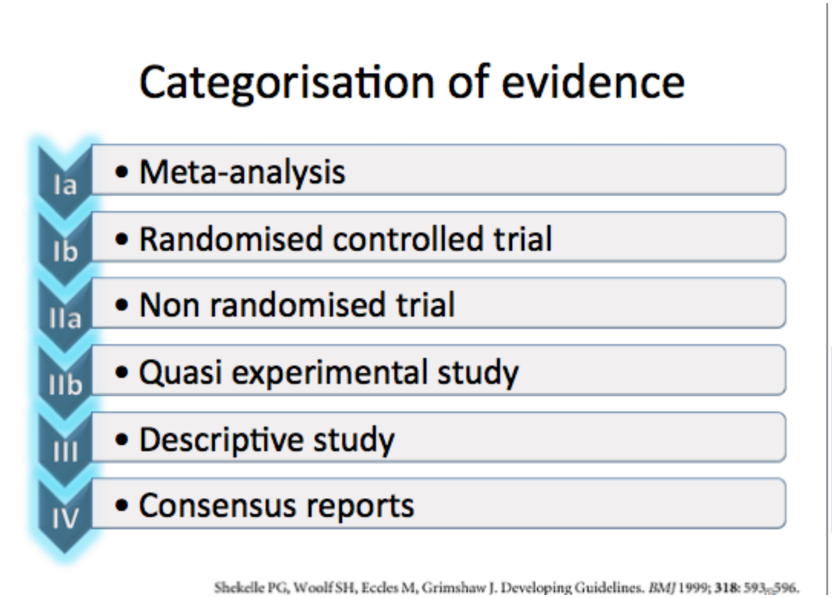
EBM involves 2 fundamental principles. It posits a hierarchy of evidence to guide clinical decision-making. Second, evidence alone is never sufficient to make a clinical decision. Decision makers are required to trade off the benefits and risks associated with alternative management strategies by considering patients' values and preferences.

Hierarchy of evidence base



- The above pyramid displays the hierarchy for treatment (and prevention) decision. Similarly, different hierarchies can be constructed for diagnostic decisions and prognostic decisions.
- Different sources provide different hierarchies: Oxford Centre for EBM provides the following system of categorising clinical evidence.
- Note that n-of-1 trials conducted on the patient for whom a treatment decision must be made if available, will occupy the highest grade.

- This hierarchy is not absolute. If treatment effects are sufficiently large and consistent, carefully conducted observational studies may provide more compelling evidence than poorly conducted RCTs.



Step 1: Identifying clinical problem

There are 5 fundamental types of clinical questions (excerpts from Guyatt's original JAMA articles):

1. **Therapy**: determining the effect of interventions on patient-important outcomes (symptoms, function, morbidity, mortality, costs). Commonly assessed using RCTs and meta-analysis.

2. **Harm**: ascertaining the effects of potentially harmful agents (including therapies from the first type of question) on patient-important outcomes. Commonly assessed using RCTs and meta-analysis.

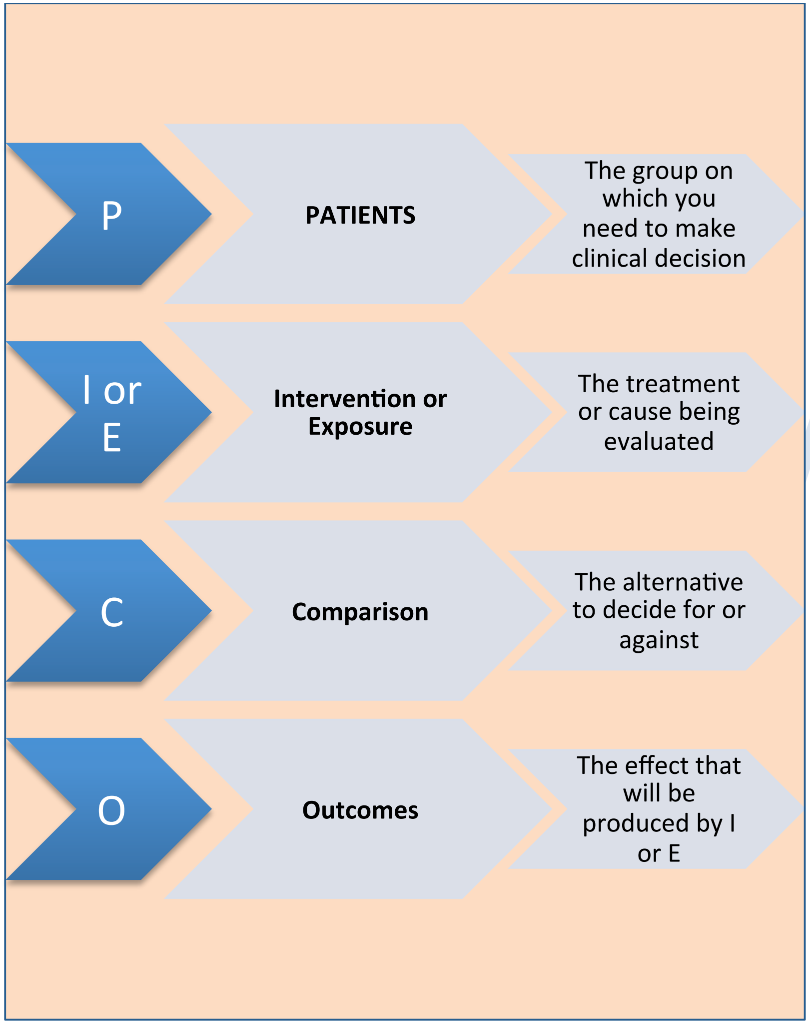
3. **Diagnosis**: establishing the power of a test to differentiate between those with and without a target condition or disease. Commonly assessed using cross-sectional studies.

4. **Prognosis**: estimating a patient's future course. Commonly assessed using cohort studies.

5. **Causation:** Apart from diagnosis and treatment, a clinician often gets involved in explaining causation to patients and carers. Establishing cause and effect relationship is often done using case-control and cohort designs, though any of the other designs mentioned previously can also be used effectively.

Step 2: Defining structured question:

In EBM, a systematic approach is used in framing the questions for which evidence is sought.



Step 3: Literature search

There are various databases in which we can search to locate the studies we need to aid our decision-making. Despite the availability of different databases, only 60-70% of scientific literature can be covered in a typical search.

Medline:

- The pre-eminent general medical database produced by the National library of Medicine
- It can be accessed via a number of different 'front ends' including PubMed and Ovid
- It includes more than 10 million citations from 4000 medical journals, about two-thirds with abstracts.
- The database starts from I966

Embase:

- Consists of 3 linked databases Excerpta Medica and 2 specialised databases for pharmacology and psychiatry

CINAHL:

- The Cumulative Index to Nursing & Allied Health (CINAHL) database specializes in literature relating to nursing and allied health professions

PsycLIT:

- It is a database produced by the American Psychological Association that covers psychological journals and books from 1887 to the present.
- It contains medical and psychiatric citations that include foreign language articles.

Cochrane Library:

- The Cochrane library consists of a collection of databases including Database of Systematic Reviews: Database of Abstracts of Reviews of Effectiveness; The Cochrane Controlled Trials Register, and the Cochrane Methodology Register.

SIGLE:

- System of information on Grey literature in Europe

Includes many difficult-to-obtain dissertations and conference abstracts

Impact factor: This is one of many quantitative tools for evaluating journals."The impact factor for a journal is calculated based on a three-year period, and can be considered to be the average number of times published papers are cited up to two years after publication. For example, the impact factor 2010 for a journal would be calculated as follows:
A = the number of times articles published in 2008-9 were cited in indexed journals during 2010.
B = the number of articles, reviews, proceedings or notes published in 2008-2009" Then impact factor 2010 = A/B
(excerpt from www.ijptonline.com).

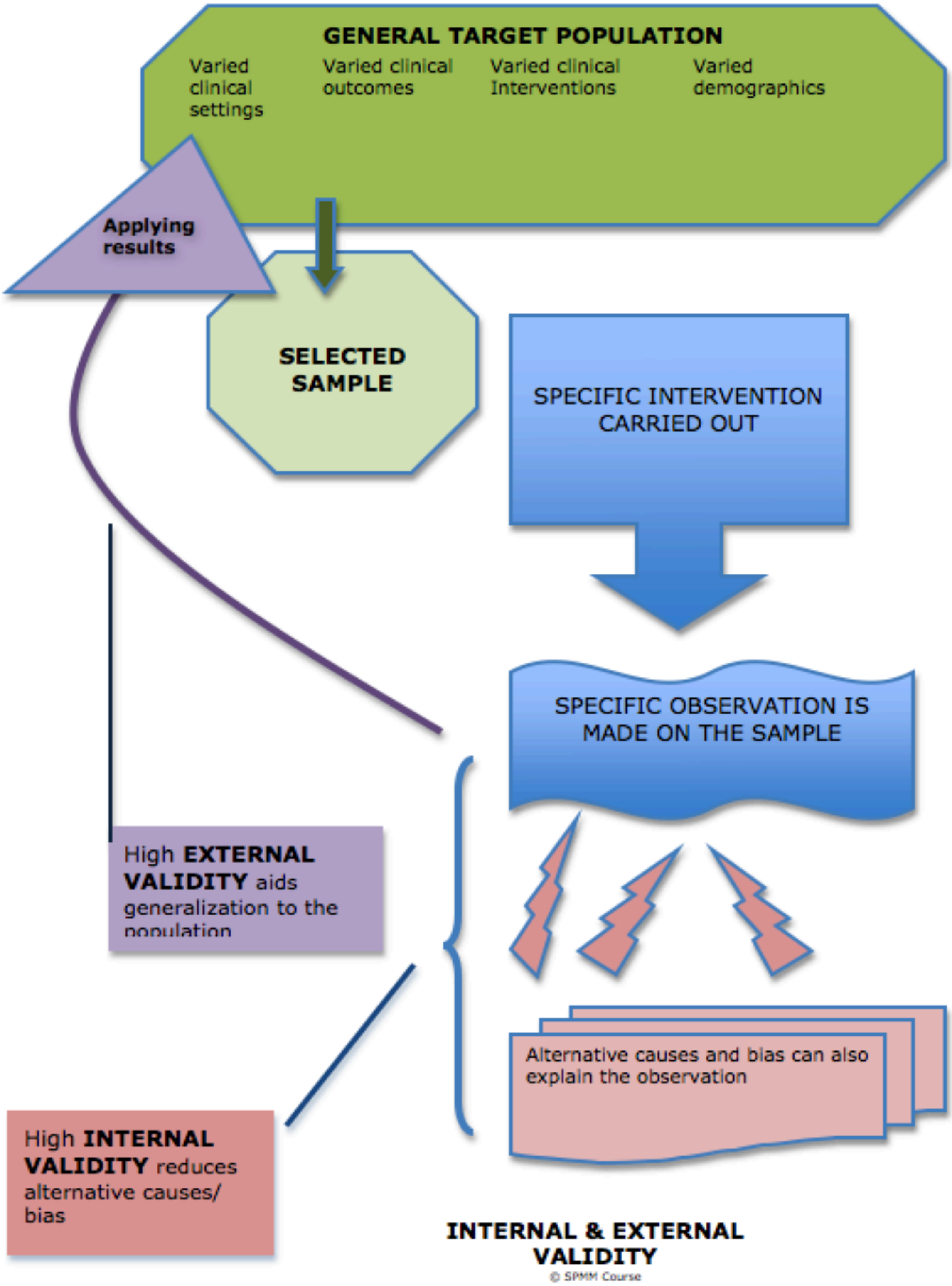
Types of reporting bias encountered when searching literature
Publication bias: Results are more apt to be published if they are significant or shown in a large sample. So many negative studies and small studies remain unpublished.
Time lag bias: Significant results are published sooner than non-significant results.
Language bias: Significant results are submitted to English-language journals, non-significant results to non–English language journals.
Database bias: Studies with significant results are more likely to be published in a journal that is indexed in a database.
Citation bias: Likelihood of article being cited depends on the significance of results and size of the trial.
Duplicate publication bias: Results of the same study appear in more than one publication.
Outcome reporting bias: Selective reporting of some study results and not reporting other analyses.

Tips for using Medline:

MEDLINE SEARCH OPERATOR	RESULTS
AND	Retrieves results that include all the search terms. By default, PUBMED utilizes AND option between two concepts. For e.g. searching (dna vitamin c), is same as searching (dna AND vitamin c)
NOT	Excludes the retrieval of terms from your search. Useful if you want to eliminate certain citations. For e.g. searching (psychosis NOT schizophrenia,) will give you psychosis related citations that do not focus on schizophrenia
OR	Retrieves results that include at least one of the search terms.
Double Quotes	Pubmed generally searches for phrases using its MeSH database. If you are using a term that is not in MeSH, then use double quotes to enable phrase search
Asterisk	To search for all terms that begin with a word, enter the word followed by an asterisk (*). Wild Card character – if you want to include all those terms that begin with schizo, then search (schizo*)
"search term"[mh] or [sh]	Medical Subject Headings (MeSH) is a controlled vocabulary of biomedical terms that is used to describe the subject of each journal article in MEDLINE. MeSH contains more than 23,000 terms. [mh[specifies Mesh Terms while [sh] specifies subheadings down the hierarchy
"name" [au]	To find the name in the author field only

Step 4: Examining the validity of evidence:

Internal validity refers to the extent that a study can be used to draw conclusions about cause and effect. High levels of bias, confounding factors and measurement errors threaten internal validity.



Random assignment of intervention improves internal validity.

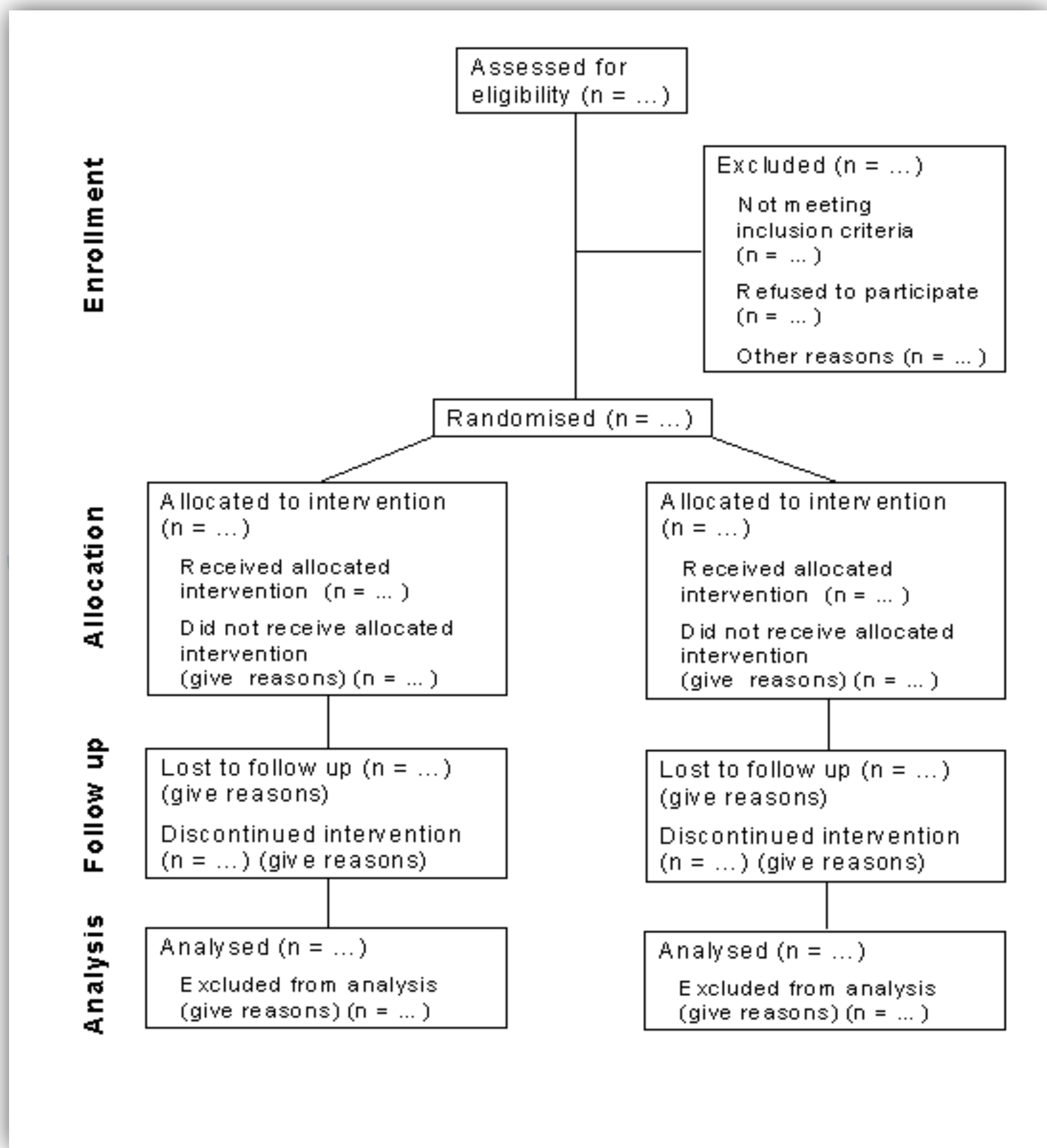
External validity refers to the extent that a study can be generalized to other places, subjects and times. It depends on pragmatic aspects of a study and reliability. Having too stringent inclusion criteria may decrease external validity though it may or may not help internal validity. In a pragmatic study where research is done in a real world setting as much as possible, certain factors such as blinding, drop outs, etc. cannot be controlled, reducing internal but increasing external validity. Random sampling improves external validity.

Steps 4 and 5 will be dealt in detail in the remaining chapters of the EBM module

Reporting conventions:

Over the years, many consensus statements have appeared to regulate reporting of various types of studies. The most popular ones are listed below.

CONSORT - CONsolidated Standards Of Reporting Trials:

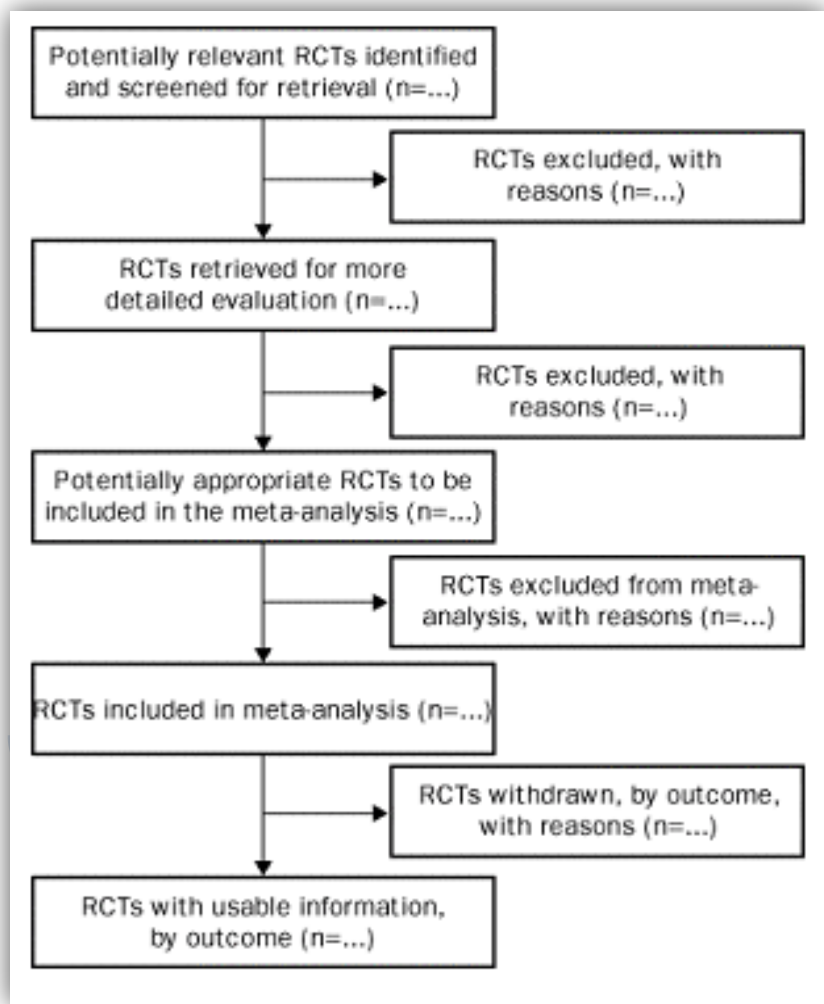


http://www.consort-statement.org/consort-statement/flow-diagram/

- CONSORT statement is an evidence-based, minimum set of recommendations for reporting RCTs.

- It offers a standard way for authors to prepare reports of trial findings, facilitating their complete and transparent reporting, and aiding their critical appraisal and interpretation

The Quality of Reporting of Meta-analyses (QUOROM) statement



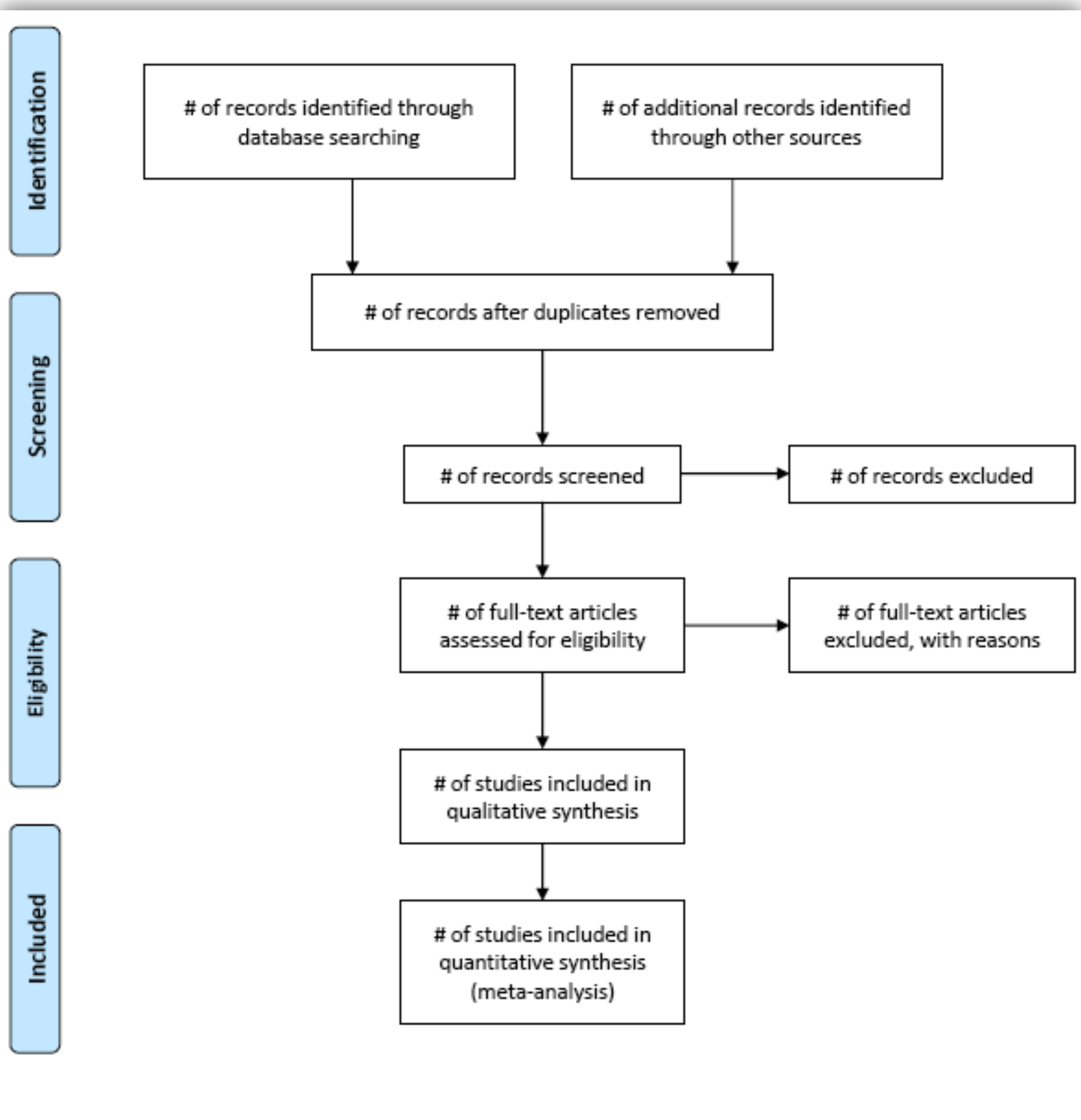
http://www.consort-statement.org/QUOROM.pdf

- QUOROM statement is an evidence-based, minimum set of recommendations for improving the quality of reporting of meta-analyses of clinical randomised controlled trials (RCTs)
- QUOROM aids authors in preparing reports of trial findings, enabling more complete and transparent reporting essential for a critical appraisal and interpretation of the data.

PRISMA - Preferred Reporting Items for Systematic Reviews and Meta-Analyses

- It is an evidence-based minimum set of items for reporting in systematic reviews and meta-analyses.
- It replaces QUOROM as of 2009

- The aim of the PRISMA Statement is to help authors improve the reporting of systematic reviews and meta-analyses. Though the main focus is on RCTs, PRISMA can also be used as a basis for reporting systematic reviews of other types of research.



From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). *P*referred *R*eporting *I*tems for *S*ystematic Reviews and *M*eta-*A*nalyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

STROBE stands for an international, collaborative initiative of epidemiologists, methodologists, statisticians, researchers and journal editors involved in the conduct and dissemination of observational studies, with the common aim of **STrengthening the Reporting of OBservational studies in Epidemiology**.

The ASSERT statement is the articulation of **A Standard for the Scientific and Ethical Review of Trials. "**It proposes a structured approach whereby research ethics committees review proposals for and monitor the conduct of, randomized controlled clinical trials. In order to ensure the ethical conduct of research involving human subjects, the ASSERT checklist comprises items that need to be addressed by investigators applying for approval to conduct a clinical trial"**.** [Retrieved from http://www.assert-statement.org/home.html]

The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) Statement

The STROBE Statement is a checklist of 22 items that we consider essential for good reporting of observational studies

Table. **The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) Statement: Checklist of Items That Should Be Addressed in Reports of Observational Studies**

Item	Item Number	Recommendation
Title and abstract	1	(a) Indicate the study's design with a commonly used term in the title or the abstract. (b) Provide in the abstract an informative and balanced summary of what was done and what was found.
Introduction		
Background/ rationale	2	Explain the scientific background and rationale for the investigation being reported.
Objectives	3	State specific objectives, including any prespecified hypotheses.
Methods		
Study design	4	Present key elements of study design early in the paper.
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection.
Participants	6	(a) Cohort study: Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up. Case–control study: Give the eligibility criteria, and the sources and methods of case ascertainment and control selection. Give the rationale for the choice of cases and controls. Cross-sectional study: Give the eligibility criteria, and the sources and methods of selection of participants. (b) Cohort study: For matched studies, give matching criteria and number of exposed and unexposed. Case–control study: For matched studies, give matching criteria and the number of controls per case.
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable.
Data sources/ measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group.
Bias	9	Describe any efforts to address potential sources of bias.
Study size	10	Explain how the study size was arrived at.
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen, and why.
Statistical methods	12	(a) Describe all statistical methods, including those used to control for confounding. (b) Describe any methods used to examine subgroups and interactions. (c) Explain how missing data were addressed. (d) Cohort study: If applicable, explain how loss to follow-up was addressed. Case–control study: If applicable, explain how matching of cases and controls was addressed. Cross-sectional study: If applicable, describe analytical methods taking account of sampling strategy. (e) Describe any sensitivity analyses.
Results		
Participants	13*	(a) Report the numbers of individuals at each stage of the study—e.g., numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analyzed. (b) Give reasons for nonparticipation at each stage. (c) Consider use of a flow diagram.
Descriptive data	14*	(a) Give characteristics of study participants (e.g., demographic, clinical, social) and information on exposures and potential confounders. (b) Indicate the number of participants with missing data for each variable of interest. (c) Cohort study: Summarize follow-up time—e.g., average and total amount.
Outcome data	15*	Cohort study: Report numbers of outcome events or summary measures over time. Case–control study: Report numbers in each exposure category or summary measures of exposure. Cross-sectional study: Report numbers of outcome events or summary measures.
Main results	16	(a) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (e.g., 95% confidence intervals). Make clear which confounders were adjusted for and why they were included. (b) Report category boundaries when continuous variables were categorized. (c) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period.
Other analyses	17	Report other analyses done—e.g., analyses of subgroups and interactions and sensitivity analyses.
Discussion		
Key results	18	Summarize key results with reference to study objectives.
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias.
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence.
Generalizability	21	Discuss the generalizability (external validity) of the study results.
Other information		
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based.

*Give such information separately for cases and controls in case–control studies, and, if applicable, for exposed and unexposed groups in cohort and cross-sectional studies.

An Explanation and Elaboration article (18–20) discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available at www.annals.org and on the Web sites of *PLoS Medicine* [www.plosmedicine.org] and *Epidemiology* [www.epidem.com]). Separate versions of the checklist for cohort, case–control, and cross-sectional studies are available on the STROBE Web site (www.strobe-statement.org).

http://www.strobe-statement.org/

The Meta-analysis Of Observational Studies in Epidemiology (MOOSE) statement:

The MOOSE is a reporting checklist for Authors, Editors, and Reviewers of Meta-analyses of *Observational Studies*.

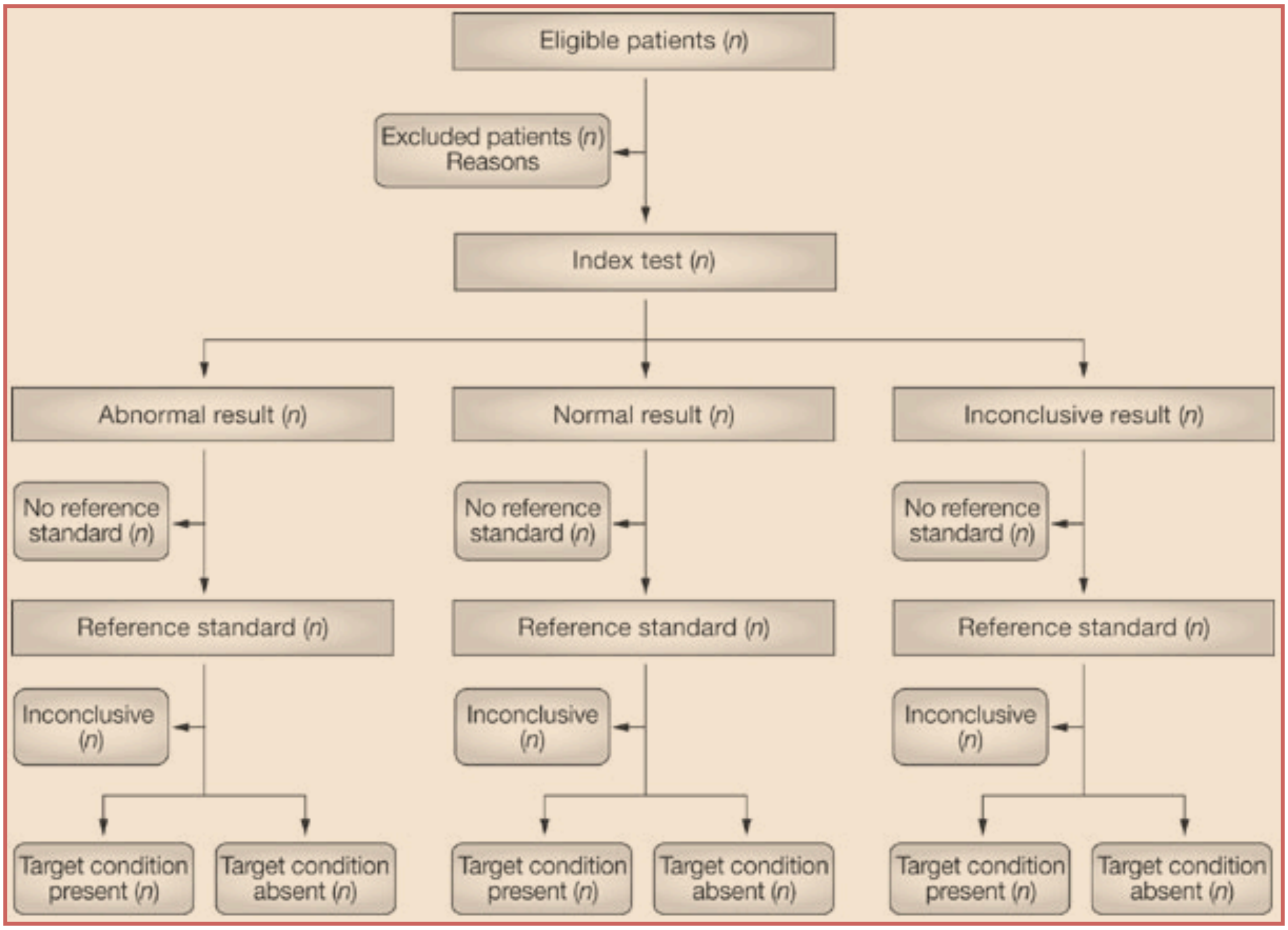
Table. A Proposed Reporting Checklist for Authors, Editors, and Reviewers of Meta-analyses of Observational Studies

Reporting of background should include	
	Problem definition
	Hypothesis statement
	Description of study outcome(s)
	Type of exposure or intervention used
	Type of study designs used
	Study population
Reporting of search strategy should include	
	Qualifications of searchers (eg, librarians and investigators)
	Search strategy, including time period included in the synthesis and keywords
	Effort to include all available studies, including contact with authors
	Databases and registries searched
	Search software used, name and version, including special features used (eg, explosion)
	Use of hand searching (eg, reference lists of obtained articles)
	List of citations located and those excluded, including justification
	Method of addressing articles published in languages other than English
	Method of handling abstracts and unpublished studies
	Description of any contact with authors
Reporting of methods should include	
	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested
	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)
	Documentation of how data were classified and coded (eg, multiple raters, blinding, and interrater reliability)
	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)
	Assessment of study quality, including blinding of quality assessors; stratification or regression on possible predictors of study results
	Assessment of heterogeneity
	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated
	Provision of appropriate tables and graphics
Reporting of results should include	
	Graphic summarizing individual study estimates and overall estimate
	Table giving descriptive information for each study included
	Results of sensitivity testing (eg, subgroup analysis)
	Indication of statistical uncertainty of findings
Reporting of discussion should include	
	Quantitative assessment of bias (eg, publication bias)
	Justification for exclusion (eg, exclusion of non–English-language citations)
	Assessment of quality of included studies
Reporting of conclusions should include	
	Consideration of alternative explanations for observed results
	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)
	Guidelines for future research
	Disclosure of funding source

Stroup, D. F., et al. (2000). Meta-analysis of observational studies in epidemiology: a proposal for reporting. *JAMA*, *283*(15), 2008-2012.

Standards for the reporting of diagnostic accuracy studies (STARD) statement

STARD offers a flowchart for the standard reporting and appraisal of studies examining the accuracy of diagnostic tests. (This STARD is different from the antidepressant trials ~ STAR*D)



http://www.stard-statement.org/